A ZOO Deep Learning Survey

Francois Chaubard
Stanford University
fchaubar@stanford.edu

Abstract

Zero-order optimization (ZOO) trains a network from loss evaluations alone, without back-propagating errors: the natural training rule for neuromorphic and analog hardware, non-differentiable systems, biologically plausible learning and memory-constrained fine-tuning. Many ZOO rules exist but are rarely compared under one budget on the same networks. We benchmark nine backprop-free training rules (plus two added later: a new combination, 1.5-INP, and DUST)—SPSA and its curvature-preconditioned variant 1.5-SPSA, random search, classic, activity-based, iterative and decorrelated node perturbation, curvature-preconditioned activity perturbation (1.5-ANP), and the forward-forward algorithm—against backpropagation on four ≈40k-parameter architectures (an MLP on MNIST, a ConvNet on CIFAR-10, a character-level LSTM and Transformer on Penn Treebank), with the same forward passes per step (backpropagation, the reference, takes one forward and one backward pass per step), one GPU-hour, the same Adam update and a probe scale tied to the learning rate. A screening sweep over five scales selects each method's scale per task; the selection is replicated with three seeds and reported at the best validation checkpoint and at the end of the hour, alongside ablations of the protocol itself. The screening's first lesson is that the step size, not the probe scale, decides: at $\varepsilon = 10^{-4}$ eight of the ten rules improve on both image tasks and the weight-space rules close about half of their gap. DUST, a token-local node-perturbation rule for Transformers that we port to all four models, has the best mean rank and the lowest summed loss (4.86, against 5.27 for 1.5-INP, which combines 1.5-ANP's estimator with the layer-wise probing of iterative node perturbation, INP, and 5.36 for INP) and leads both sequence tasks (2.47 and 2.78 bits per character on the LSTM and the Transformer, against 2.69 and 3.11 for the next rule and 1.80 and 1.84 for backpropagation); on the image tasks INP ties classic and activity-based node perturbation within seed noise (96.9 % on MNIST and 63–65 % on CIFAR-10, against 97.0 and 69.2 %), and DUST and 1.5-INP match them on MNIST but trail on CIFAR-10 (61 and 60 %). On the Transformer, identity-path credit from the speedrun that followed DUST, which measures the top-layer error exactly and sends it down the residual path, does better still (2.56 bits per character with one forward pass per step, 2.0 at 400k and 4M, where its gap to backpropagation narrows) but needs a residual network. Curvature preconditioning leaves SPSA unchanged under Adam and does not help it under SGD; 1.5-ANP is usable on the sequence models where ANP is not, but its $\alpha = 0$ control shows that the curvature weighting is not the ingredient responsible. At ten and a hundred times the parameter count (one seed) DUST and 1.5-INP stay the two best probe rules (summed loss 4.78 and 5.00 at 400k, 4.91 and 5.08 at 4M, against 5.38 and 5.52 for INP), the gap to backpropagation widens, and the step size selected at the small size has to shrink with width. Code and run logs: https://github.com/Fchaubard/zero-order-ablations.

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1 Introduction

Backpropagation [Rumelhart et al., 1986] is the default way to train neural networks, but it is not always available. Neuromorphic and analog accelerators implement a forward pass natively and may have no mechanism to transport exact error signals backwards through the substrate; systems with non-differentiable components (simulators, hardware in the loop, discrete decisions, black-box APIs) expose only a loss value; neuroscience asks which learning rules a brain could plausibly implement with local information; and fine-tuning of very large models is increasingly limited by the activation memory that backpropagation must store [Malladi et al., 2023]. In all of these settings the learner can evaluate the loss at perturbed parameters or activities but cannot differentiate it; *zero-order optimization* (ZOO) estimates a descent direction from such evaluations alone.

The ZOO literature is old and broad: random search [Rastrigin, 1963, Matyas, 1965, Schumer and Steiglitz, 1968], simultaneous perturbation stochastic approximation [Spall, 1992, 1998], Gaussian smoothing and its convergence theory [Nesterov and Spokoiny, 2017, Duchi et al., 2015, Flaxman et al., 2005], evolution strategies [Wierstra et al., 2014, Hansen et al., 2003, Salimans et al., 2017, Mania et al., 2018], node perturbation [Widrow and Lehr, 1990, Werfel et al., 2003, Fiete and Seung, 2006, Hiratani et al., 2022] and its recent activity-based and decorrelated variants [Dalm et al., 2026], curvature-preconditioned variants [Chaubard, 2025b,a], and local learning rules such as forward-forward [Hinton, 2022]. Each paper demonstrates its method on its own architecture, budget and optimizer; a practitioner who must train without backpropagation is left with the question this paper answers empirically: *which ZOO method wins, on standard small architectures, when every method is given the same budget?*

The protocol (Sections 3 and 4) gives every probe method the same forward passes per step, one GPU-hour and the same Adam update on four ≈40k-parameter models, ties the probe scale to the learning rate so that one scalar ε is screened per method, and replicates the selection with three seeds, reported at the best validation checkpoint and at the end of the hour.

Contributions. (i) A fully specified two-stage protocol under which ten training rules—backpropagation, SPSA, 1.5-SPSA, random search, four node-perturbation variants, 1.5-ANP and forward-forward—and two rules added later, the new combination 1.5-INP and DUST, are compared on four architectures with identical forward-pass, wall-clock and optimizer budgets, three seeds (two for two pairs, Table 18), and selection on validation data only. (ii) One implementation of every rule on a shared *perturbable layer* abstraction (Appendix G), so that every method drives the same models, including an LSTM and a pre-LN Transformer; the two curvature-preconditioned rules are implemented from their code repository and specification. (iii) The resulting comparison, backpropagation reference, probe-scale / step-size decoupling and estimator-alignment diagnostic (Tables 1–8 and Figures 3–8), with code, configurations and every run log released.

2 Related Work

Weight-space perturbation. SPSA [Spall, 1992, 1998] estimates a gradient from two loss evaluations at $\boldsymbol{\theta} \pm \sigma\mathbf{z}$; with Gaussian $\mathbf{z}$ it is the gradient of the Gaussian-smoothed loss [Nesterov and Spokoiny, 2017], with convex [Duchi et al., 2015] and bandit [Flaxman et al., 2005] complexity theory (survey: Liu et al., 2020), analog-VLSI origins [Jabri and Flower, 1992, Cauwenberghs, 1992, Flower and Jabri, 1992], evolution strategies [Wierstra et al., 2014, Hansen et al., 2003, Hansen, 2016] that compete with policy gradients [Salimans et al., 2017, Mania et al., 2018], relate to SGD [Zhang et al., 2017] and apply to supervised learning [Lenc et al., 2019], adaptive-moment updates [Chen et al., 2019] and directional-derivative learning [Silver et al., 2022]. MeZO [Malladi et al., 2023] fine-tunes billion-parameter language models with two-point SPSA in place; 1.5-SPSA [Chaubard, 2025b] adds a per-direction three-point curvature estimate to damp the step in sharp regions. DeepZero [Chen et al., 2024] and Zhang et al. [2024] benchmark estimators on large pretrained models; we vary the estimator *and* the architecture at a scale where every method runs to its one-hour limit, and our SPSA, 1.5-SPSA and RS are the two-point, curvature-preconditioned and best-of-K instances of this family.

Node perturbation. Node perturbation injects noise into the units and correlates the loss change with it, a local update [Widrow and Lehr, 1990, Werfel et al., 2003, 2005, Fiete and Seung, 2006] whose stability and weight diffusion Hiratani et al. [2022] analysed; Züge et al. [2023] found weight

perturbation often similar or better on temporally extended tasks, revisited here. Dalm et al. [2026] recast it as a directional derivative (*iterative* NP), proposed *activity-based* NP from the measured activity change, and added the input decorrelation of Ahmad et al. [2023], which also accelerates backpropagation [Dalm et al., 2024, Ahmad, 2024]; 1.5-ANP [Chaubard, 2025a] adds a three-point curvature preconditioner to central-difference activity perturbation; we implement all five. DUST [Dahal et al., 2026] credits each token's output jitter with the loss change at that token to pretrain Transformers; we port it to all four models, and add identity-path credit from the speedrun it started [Gülbahar et al., 2026] on the Transformer.

Local rules and benchmarks. Forward-forward [Hinton, 2022], our representative local rule, replaces the backward pass by two forward passes with positive and negative data and a local "goodness" objective per layer. Global-error broadcasting [Clark et al., 2021], feedback alignment [Lillicrap et al., 2016, Nøkland, 2016] and synthetic gradients [Jaderberg et al., 2017] still need a learned or random backward signal, and forward gradients [Baydin et al., 2022, Ren et al., 2023] a Jacobian-vector product, so they are outside the zero-order scope. Controlled comparisons exist for the feedback-alignment family [Bartunov et al., 2018, Launay et al., 2020, Sanfiz and Akrout, 2021]; we are not aware of one that puts weight perturbation, the node-perturbation family and forward-forward on the same models under one budget (unabridged: Appendix D).

3 Methods

3.1 Problem setup and the shared update rule

Let $\boldsymbol{\theta} \in \mathbb{R}^P$ be the parameters of a network with L perturbable layers and $\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N}\sum_{n=1}^{N} \ell_n(\boldsymbol{\theta})$ the mini-batch cross-entropy loss, where ℓ_n is the per-example loss (for language modelling, the per-sequence mean of the $T = 64$ next-character cross-entropies). Every method writes an estimate $\hat{\mathbf{g}} \approx \nabla_{\boldsymbol{\theta}}\mathcal{L}$ into the gradient slot of every parameter—backpropagation the exact gradient, the zero-order methods their estimator, forward-forward the gradient of its local objectives—and the same optimizer consumes it: Adam [Kingma and Ba, 2015] with $(\beta_1, \beta_2) = (0.9, 0.99)$, learning rate η and no schedule (Eq. (1), Appendix A). Every probe method has a second scalar, the probe scale σ: the standard deviation of the parameter perturbation for the weight-space methods and of the node noise for the node-perturbation family; all probes are drawn from $\mathcal{N}(\mathbf{0}, \mathbf{I})$ and scaled by σ. The protocol *ties* the two, $\eta = \sigma = \varepsilon$, so that one scalar is swept per method; Section 5.3 measures what the tie costs. Because Adam normalises $\hat{\mathbf{g}}$ by a running estimate of its own magnitude, Eq. (1) is approximately invariant to a constant rescaling of the estimator: the normalisation constants of the estimators affect the first few hundred steps and the relative weighting of estimates across steps, not the asymptotic step size, so the comparison is insensitive to the different conventions in the literature.

3.2 Optimizers

Appendix A has full definitions (Eqs. (2)–(9)) and the budget (Table 2); here, estimator and cost.

SPSA. With $K = 8$ antithetic pairs of Gaussian probes $\mathbf{z}_k$, $\hat{\mathbf{g}}_{\text{SPSA}}$ averages the central differences $[\mathcal{L}(\boldsymbol{\theta} + \sigma\mathbf{z}_k) - \mathcal{L}(\boldsymbol{\theta} - \sigma\mathbf{z}_k)]\,\mathbf{z}_k/2\sigma$ (Eq. (2)): the Gaussian-smoothing estimator of Nesterov and Spokoiny [2017] and multi-direction SPSA [Spall, 1992] with Gaussian rather than Rademacher directions, perturbed in place as in MeZO [Malladi et al., 2023]; 16 passes, no clean pass.

1.5-SPSA. 1.5-SPSA [Chaubard, 2025b] adds a clean pass (17 passes) and divides each central difference by $c_k = \max(\hat{\kappa}_k^{\alpha}, \lambda)$, $\hat{\kappa}_k$ the three-point second difference along $\mathbf{z}_k$ ($\alpha = 0.1$, $\lambda = 1$; Eq. (3)). Under Adam c_k is inert (Section 5.4), so that section also runs the published recipe (Rademacher directions, plain SGD, $\eta = \sigma$); we use Gaussian directions throughout.

Random search. Random search [Rastrigin, 1963, Matyas, 1965, Schumer and Steiglitz, 1968] evaluates the clean loss and $K = 15$ candidates $\boldsymbol{\theta} + \sigma\mathbf{z}_k$ and keeps the best, expressed for the shared update rule as a pseudo-gradient along the improving direction and zero when none improves (Eq. (4)); through Adam this only approximates the direct jump (Appendix A).

Node perturbation (NP). Classic NP [Widrow and Lehr, 1990, Werfel et al., 2003, Fiete and Seung, 2006, Hiratani et al., 2022] injects Gaussian noise $\sigma\boldsymbol{\xi}_l$ into *every* layer's pre-activations on a noisy pass and estimates layer l's gradient as the outer product of the noise-weighted per-example loss change $\delta\mathcal{L}_n\boldsymbol{\xi}_{l,n}/\sigma$ with the clean layer input (Eq. (5)); one clean and 15 noisy passes, averaged.

Activity-based NP (ANP). ANP [Dalm et al., 2026] replaces the injected noise by the measured activity difference $\delta\mathbf{a}_l = \tilde{\mathbf{a}}_l - \mathbf{a}_l$, normalised by the squared activity perturbation summed over all layers and scaled by the total number c of perturbed units (Eq. (6)); same 1+15 passes.

1.5-ANP. 1.5-ANP [Chaubard, 2025a] applies 1.5-SPSA's curvature preconditioner to central-difference activity perturbation ($K = 8$ pairs plus a clean pass, 17 passes), per *attribution unit* (an image example or a text token), with $\alpha = 0.5$ and $\lambda = 10^{-3}$ (Eq. (7)); $\alpha = 0$ gives central-difference ANP, the control that separates attribution from preconditioning (own recipe: Section 5.4); *1.5-INP* (added later) uses it at $\alpha = 0$ with INP's layer-wise probing (Appendix A).

Iterative NP (INP). INP [Dalm et al., 2026] perturbs *one* layer per pass, so the loss change is attributable to it: a directional derivative along the noise, $\lfloor 15/L \rfloor$ draws per layer (Eq. (8)).

DUST. DUST [Dahal et al., 2026], added after the protocol, jitters one group of layer outputs per draw and rewards each unit's jitter with the loss change at that unit (token or example), centred across draws; attention internals and the readout get cheap draws on cached activations (15 draws and a clean pass; Appendix A).

Identity-path credit (IPC). IPC, from the community speedrun built on DUST [Gülbahar et al., 2026] and also added later, measures each token's loss gradient at the last residual stream by central differences along a full orthonormal cycle through the final normalization and readout, hands that credit to every block along the residual path, and trains block internals by one local step from it; it needs a residual stream, so it runs on the Transformer only (one pass per step; Appendix A).

Decorrelated ANP (DANP). DANP [Dalm et al., 2026] is ANP on layer inputs decorrelated by a matrix $\mathbf{R}_l$ trained by the local rule of Ahmad et al. [2023] after each Adam step (Eq. (9)).

Forward-forward (FF). FF [Hinton, 2022] trains each block with a local goodness objective (mean of squares and threshold $\theta_{\text{FF}} = 2$ of the re-implementation of Pezeshki [2023]) on positive and negative data, forwarded as one batch with the neutral-label inputs (three N-sized passes per step on the image tasks, two on the text tasks), and never propagates an error between blocks; we report Hinton's label accumulation and a linear readout on the top block (Appendix A).

Backpropagation (reference) and forward-pass budget. BP feeds the exact gradient of one forward and one backward pass to the same Adam update. The probe methods thus spend 16 forward passes per step, the two 1.5-order rules 17, FF three or two and BP one forward plus one backward (Table 2), so steps per hour differ between methods (Appendix F), and the node-perturbation family obtains one loss difference *per example* whereas SPSA and RS obtain one batch-mean loss per pass: an asymmetry intrinsic to node perturbation that the equal-pass budget compares.

4 Experimental Protocol

Tasks and architectures. An MLP on MNIST [LeCun et al., 1998], a ConvNet on CIFAR-10 [Krizhevsky, 2009], and a character-level LSTM [Hochreiter and Schmidhuber, 1997] and Transformer [Vaswani et al., 2017] on Penn Treebank [Marcus et al., 1993, Mikolov, 2012] ($V = 50$ symbols; 5.10M / 0.40M / 0.45M characters in train / valid / test), all with ≈40k parameters (Table 3, Appendix B). Weights are Xavier-uniform [Glorot and Bengio, 2010] with zero biases (LSTM forget-gate bias 1), and the Transformer's layer normalisation [Ba et al., 2016] has no affine parameters, so every parameter lives in a perturbable layer; the last 5k training images are validation, batches are 128 images or 64 sequences of 64 characters, and there is no augmentation, dropout or weight decay.

Stage 1: ε screening. Every (task, method) pair is run once at $\varepsilon \in \{0.1, 0.01, 0.001\}$ ($\eta = \sigma = \varepsilon$, fixed throughout) for one hour of training on one NVIDIA RTX 5090, with validation and test metrics at initialisation, every 5 minutes of training and at the end (evaluation time excluded); the two 1.5-order rules were added later under the stage-2 settings. Each pair selects the ε with the lowest validation loss at any evaluation (FF on the image tasks: the highest label-accumulation validation accuracy). Because 25 of the 32 original pairs selected this grid's lower edge, $\varepsilon = 10^{-4}$ was added for every pair under the stage-2 settings, and because that value won for eight methods on both image tasks, $\varepsilon = 10^{-5}$ for those sixteen pairs, so that a selected value has a worse neighbour on each side wherever the grid allows (Table 16; five selections sit at a grid edge). On the sequence tasks 10^{-4} loses to 10^{-3} for every rule on the LSTM and for six of the ten on the Transformer, where NP, ANP, 1.5-ANP and FF select 10^{-4} (and lose again at 10^{-5}).

Stage 2: replication. The selected configuration of every pair is run with three seeds (initialisation, data order and probes), evaluated every **minute** of training time, in full fp32 arithmetic (TF32 disabled; Section 5.6 measures its effect), with checkpoints at the best validation loss. A run is stopped and marked diverged when a loss is non-finite or when its clean validation loss exceeds ten times its initial value at two consecutive evaluations, which stops collapsed networks; it also stops over-confident backpropagation runs at $\eta = 10^{-3}$ on CIFAR-10 and 10^{-2} on MNIST (Table 5), none of which is the selected reference (Appendix B: the replaced stage-1 rule). For BP, which has no probe and takes one forward and one backward pass per step (so six to ten times more steps per hour than the probe rules), the reference on each task is the learning rate with the lowest validation loss among $\eta \in \{10^{-2}, \ldots, 10^{-5}\}$ (10^{-5} on the image tasks), selected by the same rule as the probe methods and replicated with three seeds; decoupled weight decay 10^{-2} [Loshchilov and Hutter, 2019] is run as a single-seed check at $\eta \in \{10^{-3}, 10^{-4}\}$ (Section 5.2).

Reporting. The primary metric is the *best validation checkpoint*: per pair and seed, the evaluation with the lowest validation loss over the hour, whose test loss and accuracy or bits per character (loss/ $\ln 2$) are reported as the seed mean with the standard deviation in units of the last digit. The end-of-hour value measures stability; FF on the image tasks is selected and reported by label-accumulation accuracy; test data are never used for selection (Appendix B).

Ablations. (i) *Untied probe scale and step size*: the 3×3 grid $\eta, \sigma \in \{10^{-2}, 10^{-3}, 10^{-4}\}$ plus $\sigma = 0.1$ at $\eta = 10^{-4}$ per probe method and task (its $\eta = 10^{-3}$ off-diagonal for the 1.5-order rules only), screened for 30 minutes, with the best untied setting replicated for an hour with three seeds where it beats the best tied value by at least 3 % at the 30-minute checkpoint. (ii) *The 1.5-order rules under their own recipe*: plain SGD, fp64 losses, their own (η, σ) grids and $\alpha = 0$ controls (Section 5.4). (iii) *Controls under the protocol*: 1.5-ANP at the selected step size ($\eta = 10^{-4}$, Adam) with $\sigma \in \{10^{-2}, 10^{-3}\}$ decoupled, fp64 losses and $\alpha \in \{0, 0.25, 0.5, 1\}$ (Table 8), 1.5-SPSA with a non-inert factor. (iv) *Estimator alignment*: the cosine between every estimator and the true gradient of a fixed batch against the number of forward passes (Section 5.6). (v) *Resources*: steps per hour, seconds per step and peak GPU memory of every run (Appendix F). Appendix B specifies each in full and lists hardware, software, code commits and logging.

5 Results

5.1 Main comparison

Table 1 reports the best-validation-checkpoint test loss and accuracy or bits per character of every method on every task, sorted by an overall score (the sum of its four test losses, charted in Figure 1), Table 19 the same runs at the end of the hour and Table 16 the screening behind every selection. On the image tasks the node-perturbation family leads: DUST, 1.5-INP, INP, NP and ANP are within 0.01 of each other on MNIST (0.104–0.112, 96.6–97.0 %), and NP, INP and ANP within 0.05 on CIFAR-10 (1.011–1.058, 63–65 %, seed spread up to 0.06), where DANP is 0.09, DUST 0.11 and 1.5-INP 0.13 behind. The weight-space rules follow at a distance that grows with the task: SPSA and 1.5-SPSA reach 95.8–95.9 % on MNIST and 56–58 % on CIFAR-10, RS 94.9 and 53.7 %. On the sequence tasks DUST is the best probe method by a wide margin (2.47 and 2.78 bits per character on the LSTM and the Transformer); on the LSTM 1.5-INP follows (2.69, at $\varepsilon = 10^{-2}$) ahead of INP, 1.5-SPSA and NP (2.91, 2.96 and 3.02), on the Transformer 1.5-SPSA, 1.5-INP and INP (3.11–3.14) ahead of

Table 1: Main result, rows sorted by score = the sum of the four test losses. Per task: best-validation-checkpoint test loss and accuracy (%) or bits per character, mean ± s.d. over three seeds (two for the DANP and FF Transformer cells). Bold: best probe method per task. BP (one forward and one backward pass per step) is the reference; FF is reported by label-accumulation accuracy [readout in brackets] and its readout loss is not comparable, so it has no score. 1.5-INP ($\alpha = 0$, fp64 losses), DUST and IPC were added after the protocol (Section 5.4); IPC runs on the Transformer only and has no score.

	MNIST / MLP		CIFAR-10 / ConvNet		PTB / char-LSTM		PTB / char-Transformer		
Method	loss	acc. (%)	loss	acc. (%)	loss	bits/char	loss	bits/char	score
BP	0.102±0.004	97.0±0.2	0.907±0.010	69.2±0.6	1.245±0.012	1.80±0.02	1.275±0.014	1.84±0.02	3.53
DUST	0.105±0.004	96.9±0.1	1.119±0.030	60.7±1.2	**1.712±0.016**	**2.47±0.02**	**1.928±0.020**	**2.78±0.03**	4.86
1.5-INP	**0.104±0.002**	**97.0±0.1**	1.139±0.015	60.0±0.5	1.867±0.013	2.69±0.02	2.164±0.010	3.12±0.01	5.27
INP	0.107±0.004	96.9±0.2	1.055±0.015	63.5±0.5	2.017±0.055	2.91±0.08	2.176±0.022	3.14±0.03	5.36
NP	0.107±0.002	96.7±0.1	**1.011±0.061**	**65.0±2.6**	2.092±0.067	3.02±0.10	2.328±0.011	3.36±0.02	5.54
1.5-SPSA	0.140±0.008	95.9±0.2	1.215±0.048	56.7±1.9	2.054±0.008	2.96±0.01	2.155±0.026	3.11±0.04	5.56
SPSA	0.142±0.005	95.8±0.2	1.200±0.051	57.8±2.0	2.141±0.082	3.09±0.12	2.200±0.017	3.17±0.02	5.68
RS	0.168±0.012	94.9±0.3	1.292±0.056	53.7±2.2	2.229±0.092	3.22±0.13	2.276±0.027	3.28±0.04	5.97
1.5-ANP	0.159±0.007	95.7±0.4	1.532±0.041	46.2±1.0	2.167±0.021	3.13±0.03	2.566±0.085	3.70±0.12	6.42
DANP	0.211±0.002	94.0±0.1	1.101±0.018	63.0±1.1	2.635±0.040	3.80±0.06	2.578±0.092	3.72±0.13	6.53
ANP	0.112±0.003	96.6±0.1	1.058±0.025	63.3±1.0	2.885±0.082	4.16±0.12	2.735±0.039	3.95±0.06	6.79
FF	2.335±0.058	59.7±29.5 [11.0]	2.007±0.036	33.9±0.3 [26.9]	2.618±0.035	3.78±0.05	2.932±0.016	4.23±0.02	–
IPC	–	–	–	–	–	–	1.777±0.027	2.56±0.04	–

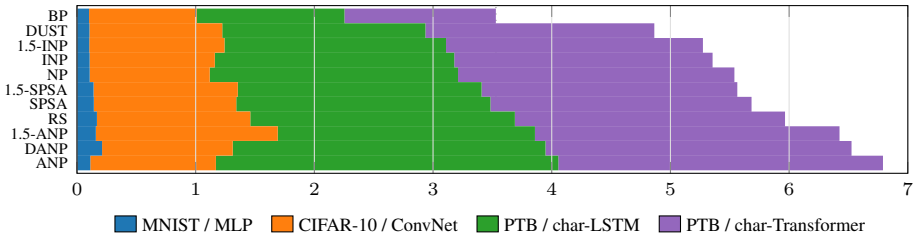


Figure 1: The score of Table 1 as a chart: best-checkpoint test loss summed over the four tasks (lower is better), with the per-task losses stacked (dotted line: BP); FF omitted (readout loss not comparable).

SPSA (3.17), and ANP is last on both (4.16 and 3.95). IPC, Transformer only, beats every rule there but BP: 1.777 (2.56 bits per character), 0.15 below DUST and a quarter of the way to BP, with one pass per step (308k steps in the hour, DUST 78k); it passes DUST's one-hour loss in minute 6 and then flattens. Across the four tasks DUST has the best mean rank among the ten probe methods at the best checkpoint (2.25, bootstrap interval 1.75–3.0), ahead of INP and 1.5-INP (3.0), NP (4.25), 1.5-SPSA (5.0), SPSA (6.25), ANP (7.0), RS and DANP (8.0) and 1.5-ANP (8.25), and the lowest score (4.86, against 5.27 for 1.5-INP and 5.36 for INP). Against backpropagation at its own best learning rate (97.0 and 69.2 %, Section 5.2) the best probe rule is 0.1 points behind on MNIST and 4 points behind on CIFAR-10, and on the LSTM and the Transformer the gap is 0.7 and 0.9 bits per character, a factor 1.4–1.5 in test loss. The end-of-hour view (Table 19) promotes NP and 1.5-SPSA, which drift least, keeps DUST's score the lowest (4.96; 1.5-INP 5.39), and on the sequence tasks changes, among the probe rules, only ANP, 1.5-ANP and DANP, which drift. Figures 3–5 (Appendix C) show the curves at the three sizes: on the image tasks BP peaks within five minutes and then grows over-confident, the weight-space rules and 1.5-SPSA have their best checkpoint in the last fifteen minutes, and on the sequence tasks every rule but ANP, 1.5-ANP and DANP is still descending or flat at minute 60.

5.2 The backpropagation reference

With Adam and no schedule, backpropagation completes 2.7–4.3M steps on MNIST (seed spread) and 2.0M on CIFAR-10 in the hour and overfits within minutes: at $\eta = 10^{-3}$ its CIFAR-10 validation loss never returns below its initial value, and at the selected reference ($\eta = 10^{-5}$ on MNIST, 10^{-4} on CIFAR-10) the best checkpoint sits at minute 1–5 (97.0 and 69.2 %) and the hour ends at 96.7 and 65.6 % (Table 5). On the sequence tasks $\eta = 10^{-3}$ is best and BP is still improving at minute 60 (1.80 and 1.84 bits per character). Decoupled weight decay (single seed) adds 0.3–0.5 points on MNIST at $\eta = 10^{-3}$ (0.1–0.2 at 10^{-4}) and removes its drift, and changes little elsewhere (Table 5).

5.3 Probe scale and step size

The tied screening (Table 16) was decisive but for the wrong knob: $\varepsilon \geq 0.01$ collapses most probe runs, 10^{-3} loses none, and 10^{-4} improves eight of the ten methods on both image tasks once more (SPSA from 0.285 to 0.132 in validation loss on MNIST, 1.62 to 1.27 on CIFAR-10), while 10^{-5} is worse for all but four pairs (BP, NP and ANP on MNIST, 1.5-ANP on CIFAR-10, where it ties or wins by a margin within seed noise). The untied grid (Table 6, Appendix C) attributes most of this to the step size: at fixed $\eta = 10^{-4}$ the probe scale ($\sigma \leq 10^{-2}$) changes the 30-minute loss of SPSA, 1.5-SPSA, RS, NP and INP by at most 0.03 on MNIST and 0.09 on CIFAR-10, while ANP and DANP are σ-sensitive on CIFAR-10 (0.14 and 0.21) and 1.5-ANP everywhere; on the image tasks $\eta = 10^{-2}$ loses to 10^{-4} for every rule but DANP on CIFAR-10. A noisy estimate wants a small Adam step and many of them. On the sequence tasks $\eta = 10^{-4}$ loses to 10^{-3} for every rule on the LSTM (12–34k steps per hour) and for six of ten on the Transformer (Table 16). The best untied cell beats the best tied one by more than 3 % only for 1.5-ANP on both image tasks (24 and 17 %, at $\sigma \geq 10^{-3}$), DANP on the Transformer (6 %) and ANP on MNIST (4 %); one-hour replications confirm them (Table 9).

5.4 The 1.5-order rules under their own recipe

Table 4 (Appendix C) runs both rules as published—plain SGD, fp64 losses, their own (η, σ) and probe-count grids—and Table 8 (Appendix C) runs 1.5-ANP under the protocol's optimizer at the step size the screening selected, with its probe scale freed. Neither recipe beats the protocol: 1.5-SPSA with SGD and 64 pairs per step reaches 0.176 on MNIST and 1.336 on CIFAR-10 against 0.160 and 1.313 for its Adam run at the same 30 minutes, and its $\alpha = 0$ control (plain SPSA with SGD) is as good or better on every task. The curvature factor of 1.5-SPSA is thus inert under Adam, which removes its nearly common scale, and slightly harmful under SGD. For 1.5-ANP, Adam at $\eta = 10^{-4}$ with fp64 losses and $\sigma = 10^{-2}$ improves 0.159 to 0.132 on MNIST and 1.53 to 1.28 on CIFAR-10 (its tied selection was limited by a probe scale too small for a second difference in fp32), but at $\sigma = 10^{-3}$ its $\alpha = 0$ control is better still (0.112 and 1.120 against 0.133 and 1.307 with $\alpha = 0.5$), equals ANP on MNIST and comes within 0.06 of it on CIFAR-10; $\alpha = 0.25$ lies in between and $\alpha = 1$ is harmful (0.204 and 1.98). The same holds at the tied $\varepsilon = 10^{-3}$ (Table 7), on the sequence models (Table 8) and, under its own SGD recipe, on two of the four tasks (Table 4). What makes 1.5-ANP usable on the sequence models, where it beats ANP by 1.0 and 0.25 bits per character, is therefore not the curvature exponent; its other differences to ANP (central differences, per-unit normalisation, token-local credit assignment) are not separated by any run here. A second combination, 1.5-ANP's estimator with INP's one-layer-at-a-time probing ("1.5-INP", Appendix A), was added after the protocol, as was DUST; it enters Table 1 at $\alpha = 0$ (fp64 losses, the same confound; ε bracketed in Table 14), leads INP through the LSTM (1.867 against 2.017) and loses on CIFAR-10 (1.139 against 1.055); untied σ and $\alpha = 0.5$ are in Table 13.

5.5 Model size

Figure 2 repeats the protocol at ten and a hundred times the parameter count (widths scaled, depth unchanged; Table 10): every (task, method) pair reruns for one hour, one seed, at the ε selected at 40k, and every pair that stalled or collapsed there was re-screened at $\varepsilon/10$ (and $/100$ on MNIST at 4M), the lowest validation loss deciding. At 400k the node rules and BP hold or improve (DUST 4.86 to 4.78, 1.5-INP 5.27 to 5.00, INP 5.36 to 5.38, NP 5.54 to 5.47, BP 3.53 to 3.04; summed loss, Figure 2) while the weight-space rules lose 0.05–0.23. At 4M most probe rules lose 0.1–0.6 against 40k (INP 5.52, SPSA 5.84, NP 5.91, RS 6.55), DANP improves to 5.88, and BP keeps improving (2.98): the gap widens, and DUST and 1.5-INP stay the two best probe rules (4.91 and 5.08). They are the two best on both sequence tasks at every size (DUST first on the LSTM, 1.5-INP on the Transformer at 400k and 4M, 1.81 against 1.87 at 4M, with every other rule above 2.13) and trail the node rules on CIFAR-10. IPC (Transformer only; $\varepsilon = 10^{-4}, 10^{-3}, 10^{-2}$ at each size; the larger runs stopped at minute 58 of 60) improves with size: 1.78, 1.39 and 1.37 at 40k, 400k and 4M against BP's 1.28, 1.06 and 1.07, so its gap to BP narrows from 0.72 to 0.44 bits per character, at the smallest ε tried at both larger sizes. Unlike the other rules, which were re-screened only where they stalled, both were run at the 40k ε and a decade below in every cell (two decades on MNIST at 4M, none on CIFAR-10 at 400k); at the 40k ε alone they still lead INP at 400k (5.09 and 5.29) but not at 4M (5.81 and 5.57), where their sequence-model runs plateau like everyone else's. The step size matters more than the estimator: at the ε selected at 40k every probe rule on the 4M Transformer plateaued from minute

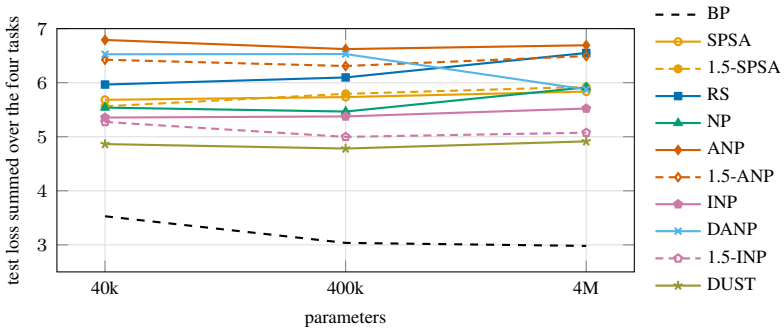


Figure 2: Test loss summed over the four tasks against model size (log scale), one seed per point, ε selected at 40k and re-screened a decade or two down where a run stalled (1.5-INP and DUST: a decade down in every cell); the 40k points are the seed means of Table 1; FF omitted, as in Figure 1.

10 at 2.5–2.9; one decade down SPSA, INP and 1.5-SPSA reach 2.14–2.17, as good as at 40k, one decade up they diverge, and plain SGD or 128 passes at the old ε do not help. The 4M LSTM is budget-limited (16–33k steps an hour). Alignment does degrade with width (Table 25: tenfold for the weight-space rules, two- to sixfold for NP, INP and DANP). Figure 7 shows the per-task curves. In two follow-ups, random search with direct jumps stays within 0.04 of the protocol's version (Table 11) and four times more pairs do not rescue 1.5-ANP's curvature term (Table 12).

5.6 Estimator quality

Figure 8 (Appendix C) measures each estimator's cosine with the true gradient against the number of forward passes: on the MLP the node rules get 0.4–0.6 from the protocol's 16 passes and the weight-space rules 0.01; on the ConvNet every estimator is below 0.1 at 16 passes; on the sequence models none exceeds 0.26 at 1024. 1.5-SPSA is indistinguishable from SPSA, and only 1.5-ANP's second difference degrades as σ shrinks in fp32 (Table 24), which is why it needs $\sigma \geq 10^{-3}$.

5.7 Analysis

Which zero-order method wins? DUST, if a single rule must be chosen: it has the lowest score at the best checkpoint and at the end of the hour (4.86 and 4.96; 1.5-INP 5.27 and 5.39, INP 5.36 and 5.58) and at 400k and 4M, the best mean rank, and leads both sequence tasks by 0.2–0.3 bits per character, but is 0.11 behind on CIFAR-10, where ANP, NP and INP are indistinguishable and DANP is the fastest. It differs from INP by rewarding each unit's jitter with the loss change at its own token rather than the sequence mean, centring the reward across draws, and giving attention internals and the readout cheap draws on cached activations; on the image models, where both credit an example's own loss, only the last two differ. The gap to BP is 0.1 points on MNIST, 4 points on CIFAR-10 and 0.7–0.9 bits per character on the sequence tasks. IPC does better on the Transformer by using its residual stream, so it does not apply to the MLP, ConvNet or LSTM.

Weight probes versus node probes. SPSA and RS perturb all $P \approx 40$k weights and obtain one scalar loss difference per probe, so their 16 passes yield a direction with cosine 0.01 to the gradient on the MLP (Figure 8), whereas the node rules obtain a loss difference per example and assign credit per unit (cosine 0.4–0.6); at the right step size the gap is 1 point on MNIST and 7 points on CIFAR-10, and it closes on the sequence models, where SPSA ends within 0.18 and 0.03 bits per character of INP and ahead of NP on the Transformer, as Züge et al. [2023] found for temporally extended tasks.

Curvature preconditioning. 1.5-SPSA reproduces SPSA to within 0.05 in loss on three tasks and is 0.09 better on the LSTM, where SPSA's seed spread is 0.08, at the price of the clean pass (13–15 % fewer steps per hour on the image tasks): with $\alpha = 0.1$ and $\lambda = 1$ the per-direction factor is nearly constant and Adam removes the remaining common scalar, and Section 5.4 shows the same under plain SGD. 1.5-ANP is one point behind ANP on MNIST and 17 points behind on CIFAR-10 under the tied protocol, because a second difference at $\sigma \leq 10^{-4}$ sits at the fp32 rounding floor; with $\sigma = 10^{-2}$ and fp64 losses it recovers to within 0.03 of ANP on MNIST, and its $\alpha = 0$ control matches ANP on

MNIST and comes within 0.06 on CIFAR-10. On the sequence models its token-local attribution repairs ANP's failure (3.13 versus 4.16 bits per character on the LSTM, 3.70 versus 3.95 on the Transformer) without reaching INP or SPSA.

Activity-based, iterative and decorrelated variants. The measured activity difference (ANP) is worth nothing over NP at the right step size (within seed noise on both image tasks), but on the recurrent and attention models its normalisation by a squared activity difference summed over 25k–48k unit-positions, whose perturbations propagate through shared weights, produces updates of erratic size (the spikes in Figure 3) and the worst probe result on both tasks. INP's one-layer-at-a-time attribution removes all cross-layer noise and is never worse than fourth on any task. Decorrelation helps only where Dalm et al. [2026] reported it: DANP is the fastest rule on CIFAR-10 (best checkpoint at minute 5, 63.0 %) but drifts to 1.36 by minute 60, is behind on MNIST (94.0 %) and on the sequence tasks (3.80 and 3.72 bits per character), and was the screening's most fragile rule. A lower η_R does not remove the drift at 40k but helps at 4M (0.120 on MNIST, 0.831 on CIFAR-10 against 0.157 and 1.025; Table 15).

Sequence models and forward-forward. Against count models on the training text (unigram 4.32 bits/char, bigram 3.31, trigram 2.61; Appendix G) the zero-order rules learn roughly bigram statistics in an hour—eight rules beat the bigram on the LSTM (2.47–3.22) and six on the Transformer (2.78–3.28), only DUST (LSTM) and IPC (Transformer, 2.56) the trigram—and BP (1.80 and 1.84) a real character model; nearly every probe curve is still descending at minute 60, so part of the gap is budget, but at 12–34k steps per hour closing it would take days. FF, not a probe method, trains fast but its two inference rules disagree: on MNIST its selected $\varepsilon = 0.1$ is seed-fragile (label accumulation 59.7(295) % over three seeds against 93.1(3) % at $\varepsilon = 0.01$, Appendix C) and the linear readout collapses to 11–14 %; on CIFAR-10 both numbers are 27–34 %, and on the sequence tasks FF is third-worst on the LSTM and worst on the Transformer (3.78 and 4.23 bits per character; Appendix C).

6 Discussion and Limitations

What the final-loss protocol measures. Under a fixed Adam step with no schedule every probe run at $\varepsilon \geq 10^{-3}$ peaks within minutes and then drifts: once the estimate's signal is exhausted the parameters random-walk with step η and the network grows over-confident. The loss after a fixed hour therefore ranks methods by where their random walk settles, and selecting ε by it produced degenerate choices in stage 1; hence stage 2 reports the best validation checkpoint as primary and the final value for stability; a learning-rate schedule would be the other remedy.

Screening, budget and scale. The selection rests on one seed per ε, so the error bars exclude selection noise; the models are small, and Section 5.5 shows the weight-space rules losing ground with P. Wall-clock comparisons reward implementations that batch probes well; the equal-pass view (Figure 6) is the implementation-independent comparison. Every departure from a method's source is stated where it is defined (Appendix C).

7 Conclusion

We compared nine backprop-free training rules, three added later (1.5-INP, DUST and identity-path credit) and backpropagation on four ≈40k-parameter architectures under one equal-budget protocol. The screening sweep's first result is methodological: with Adam the step size, not the probe scale, decides how far a noisy estimator gets in an hour, and at $\eta = 10^{-4}$ the weight-space rules close about half of their gap on the image tasks. If one zero-order rule must be chosen it is DUST: it has the lowest summed loss and leads both language models by 0.2–0.3 bits per character but trails on CIFAR-10, followed by iterative node perturbation and 1.5-INP; curvature preconditioning adds nothing to SPSA under Adam or SGD, and the sequence-model gains of 1.5-ANP over ANP survive setting its curvature exponent to zero. On the language models the best zero-order rule is still 0.7 and 0.9 bits per character behind backpropagation after an hour; identity-path credit, which exploits the Transformer's residual stream, cuts the Transformer's gap from 0.9 to 0.7 bits, and to 0.4 at 4M. At ten and a hundred times the size DUST and 1.5-INP remain the two best probe rules once the step size is re-selected, and the gap to backpropagation widens.

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A Optimizer definitions

This appendix gives the shared update rule and the full definition of every rule summarised in Section 3.2; Table 2 lists the forward-pass budget of every method per step.

Shared update rule. Every estimate $\hat{\mathbf{g}}$ is consumed by Adam [Kingma and Ba, 2015] with $(\beta_1, \beta_2) = (0.9, 0.99)$, learning rate η and no schedule:

$$\mathbf{m} \leftarrow \beta_1 \mathbf{m} + (1-\beta_1)\,\hat{\mathbf{g}}, \qquad \mathbf{v} \leftarrow \beta_2 \mathbf{v} + (1-\beta_2)\,\hat{\mathbf{g}}^2, \qquad \boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\,\frac{\hat{\mathbf{m}}}{\sqrt{\hat{\mathbf{v}}} + 10^{-8}}. \tag{1}$$

Adam's constants. $(\beta_1, \beta_2) = (0.9, 0.99)$ is a protocol choice made before any run (the usual $\beta_2 = 0.999$ would average the second moment over a thousand steps, several seconds of these runs); it is shared by every method and is not tuned.

SPSA (central difference, Gaussian probes). With $K = 8$ antithetic pairs of probes $\mathbf{z}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_P)$,

$$\hat{\mathbf{g}}_{\text{SPSA}} = \frac{1}{K}\sum_{k=1}^{K} \frac{\mathcal{L}(\boldsymbol{\theta} + \sigma \mathbf{z}_k) - \mathcal{L}(\boldsymbol{\theta} - \sigma \mathbf{z}_k)}{2\sigma}\,\mathbf{z}_k, \tag{2}$$

the Gaussian-smoothing estimator of Nesterov and Spokoiny [2017] and the multi-direction form of SPSA [Spall, 1992] with Gaussian rather than Rademacher directions. Each pair costs two forward passes on the batch-mean loss, so the 16-pass budget buys eight directions. Parameters are perturbed in place and restored exactly from a flat copy, as in MeZO [Malladi et al., 2023]; no clean pass is spent, and the training loss logged for SPSA is the mean of the 16 probe losses.

1.5-SPSA. 1.5-SPSA [Chaubard, 2025b] spends one clean pass on $\mathcal{L}_0 = \mathcal{L}(\boldsymbol{\theta})$ and uses the three-point second difference along every direction to damp the step where the loss is sharp:

$$\begin{aligned} \hat{\mathbf{g}}_{\text{1.5-SPSA}} &= \frac{1}{K}\sum_{k=1}^{K} \frac{\mathcal{L}(\boldsymbol{\theta} + \sigma \mathbf{z}_k) - \mathcal{L}(\boldsymbol{\theta} - \sigma \mathbf{z}_k)}{2\sigma\, c_k}\,\mathbf{z}_k, \\ c_k &= \max\!\big(\hat{\kappa}_k^{\alpha}, \lambda\big), \qquad \hat{\kappa}_k = \frac{\lvert \mathcal{L}(\boldsymbol{\theta} + \sigma \mathbf{z}_k) - 2\mathcal{L}_0 + \mathcal{L}(\boldsymbol{\theta} - \sigma \mathbf{z}_k)\rvert}{\sigma^2}, \end{aligned} \tag{3}$$

with the saturating exponent $\alpha = 0.1$ and floor $\lambda = 1$ of the published code. With $K = 8$ pairs the step costs 17 passes. The published method uses Rademacher directions and a plain SGD step

$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\,\hat{\mathbf{g}}$ with $\eta = \sigma$ tied, i.e. $\Delta\boldsymbol{\theta} = -\frac{1}{2K}\sum_k \frac{\Delta\mathcal{L}_k}{c_k}\mathbf{z}_k$, many directions per step and few steps [Chaubard, 2025b]. Under the protocol the estimate goes to Adam like every other method, which makes c_k inert (Section 5.6); Section 5.4 therefore also runs the rule under its own recipe. We use Gaussian directions, like SPSA, throughout.

Random search. Random search [Rastrigin, 1963, Matyas, 1965, Schumer and Steiglitz, 1968] evaluates the clean loss $\mathcal{L}_0$ and $K = 15$ candidates $\boldsymbol{\theta} + \sigma\mathbf{z}_k$ and keeps the best, $k^* = \arg\min_k \mathcal{L}(\boldsymbol{\theta} + \sigma\mathbf{z}_k)$. To fit the shared update rule it is expressed through the pseudo-gradient

$$\hat{\mathbf{g}}_{\text{RS}} = \begin{cases} \dfrac{\mathcal{L}_{k^*} - \mathcal{L}_0}{\sigma}\,\mathbf{z}_{k^*} & \text{if } \mathcal{L}_{k^*} < \mathcal{L}_0, \\ \mathbf{0} & \text{otherwise,} \end{cases} \tag{4}$$

which points along the improving direction and is zero when no candidate improves. Through Adam this is only an approximation of the jump $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \sigma\mathbf{z}_{k^*}$: the first step moves by $\approx \eta\,\mathrm{sign}(\mathbf{z}_{k^*})$ per coordinate, later steps are bent by momentum, and rejected steps still move the parameters along the decaying momentum; the direct-jump variant ("jumping" random search without Adam) is compared in Table 11.

Node perturbation (NP). Classic node perturbation [Widrow and Lehr, 1990, Werfel et al., 2003, Fiete and Seung, 2006, Hiratani et al., 2022] perturbs the units rather than the weights. For layer l with input $\mathbf{h}_{l-1}$, pre-activation $\mathbf{a}_l = \mathbf{W}_l\mathbf{h}_{l-1} + \mathbf{b}_l$ and post-activation $\mathbf{h}_l = f(\mathbf{a}_l)$, a noisy pass injects Gaussian noise into *every* layer and propagates it, $\tilde{\mathbf{a}}_l = \mathbf{W}_l\tilde{\mathbf{h}}_{l-1} + \mathbf{b}_l + \sigma\boldsymbol{\xi}_l$ with $\boldsymbol{\xi}_l \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$. Writing $\delta\mathcal{L}_n = \tilde{\ell}_n - \ell_n$ for the change of the per-example loss, the estimator for layer l is

$$\hat{\mathbf{g}}^{\text{NP}}_{\mathbf{W}_l} = \frac{1}{N}\sum_{n=1}^{N} \frac{\delta\mathcal{L}_n}{\sigma}\,\boldsymbol{\xi}_{l,n}\,\mathbf{h}_{l-1,n}^{\top}, \qquad \hat{\mathbf{g}}^{\text{NP}}_{\mathbf{b}_l} = \frac{1}{N}\sum_{n=1}^{N} \frac{\delta\mathcal{L}_n}{\sigma}\,\boldsymbol{\xi}_{l,n}, \tag{5}$$

the outer product of the noise-weighted loss change with the *clean* layer input (for convolutions, the matching correlation over unfolded patches; for embeddings, the one-hot input), averaged over 15 noisy passes. $\delta\mathcal{L}_n\boldsymbol{\xi}_l/\sigma$ equals $\delta\mathcal{L}_n\mathbf{v}_l/\sigma^2$ with $\mathbf{v}_l = \sigma\boldsymbol{\xi}_l$, the form of Dalm et al. [2026]. For a layer applied several times per forward (the LSTM gate layer, once per time step) the same $\delta\mathcal{L}_n$ weights every call.

Activity-based node perturbation (ANP). Because injected noise propagates, the actual perturbation of layer l is the measured activity difference $\delta\mathbf{a}_l = \tilde{\mathbf{a}}_l - \mathbf{a}_l$ rather than $\sigma\boldsymbol{\xi}_l$. Activity-based NP [Dalm et al., 2026] uses this difference and normalises by the norm of the total activity perturbation across all layers:

$$\hat{\mathbf{g}}^{\text{ANP}}_{\mathbf{W}_l} = \frac{1}{N}\sum_{n=1}^{N} c\,\frac{\delta\mathcal{L}_n}{\|\delta\mathbf{a}_n\|^2}\,\delta\mathbf{a}_{l,n}\,\mathbf{h}_{l-1,n}^{\top}, \qquad \|\delta\mathbf{a}_n\|^2 = \sum_{l'=1}^{L}\|\delta\mathbf{a}_{l',n}\|^2, \tag{6}$$

with $c = \sum_l N_l$ the total number of perturbed units (all layers, all time steps or positions; the input is not counted), following the authors' reference implementation. For a single perturbed unit it reduces to Eq. (5).

1.5-ANP. 1.5-ANP [Chaubard, 2025a] combines activity-based perturbation with antithetic pairs and the three-point curvature preconditioner of 1.5-SPSA, applied per *attribution unit* u: an example on the image tasks and a (sequence, position) token on the language-modelling tasks ("token-local" attribution, with the LSTM gate layer called once per position). Each of $K = 8$ pairs injects $\pm\sigma\boldsymbol{\xi}$ into every layer; with $\mathbf{d}_u = \mathbf{a}_u^+ - \mathbf{a}_u^-$ the realized central activity difference over the N_u perturbed nodes of u and $\ell_u^0, \ell_u^\pm$ the clean and perturbed losses of u,

$$\hat{c}_u = \frac{4N_u\,|\ell_u^+ - 2\ell_u^0 + \ell_u^-|}{\|\mathbf{d}_u\|^2}, \qquad \beta_u = \frac{N_u\,(\ell_u^+ - \ell_u^-)}{\|\mathbf{d}_u\|^2\,(\hat{c}_u + \lambda)^{\alpha}}, \qquad \hat{\mathbf{g}}^{\text{1.5-ANP}}_{\mathbf{a}_u} = \frac{1}{K}\sum_{k=1}^{K}\beta_u^{(k)}\mathbf{d}_u^{(k)}, \tag{7}$$

with $\lambda = 10^{-3}$ (10^{-4} on the LSTM in the specification) and $\alpha = 0.5$; the weight update is the batch mean of $\hat{\mathbf{g}}_{\mathbf{a}_u}\mathbf{h}_u^{\top}$ as in Eq. (6), and N_u counts the perturbed nodes of one unit (90 on the MLP, 402 per LSTM time step, 754 per Transformer token). With $\alpha = 0$ the rule is central-difference ANP, the

control that separates attribution from preconditioning. The step costs 17 passes. *1.5-INP* keeps this estimator but perturbs one layer per pair, as INP does: pair k of step t perturbs layer $(8t + k) \bmod L$, so each step touches eight layers (all of them when $L \leq 8$, eight of the Transformer's 15), the loss difference of a unit is attributed to that layer's N_u nodes only, and the update is otherwise Eq. (7); it costs the same 17 passes and runs with fp64 losses at the ε of a bracket around INP's (Table 14). The specification uses plain SGD with a learning rate 20–50 times σ and damps the deviation of the perturbed recurrent state by one half at every time step; the protocol runs tie $\eta = \sigma$ under Adam without damping, and Section 5.4 runs the rule under its own recipe with both (the settings its own grid selects have η/σ between 0.5 and 5, Table 4, not the specification's 20–50). The three-point second differences that define $\hat{c}_u$ are, at $\sigma = 10^{-3}$, of the order of the fp32 rounding of the loss, so the own-recipe runs evaluate the losses in fp64 (Section 5.6).

Iterative node perturbation (INP). Iterative NP [Dalm et al., 2026] perturbs *one* layer per noisy pass, so that the loss change $\delta\mathcal{L}_n^{(l)}$ is attributable to that layer and the estimator is a directional derivative along the injected noise $\mathbf{v}_{l,n} = \sigma\boldsymbol{\xi}_{l,n}$,

$$\hat{\mathbf{g}}_{\mathbf{W}_l}^{\text{INP}} = \frac{1}{N}\sum_{n=1}^{N} N_l \,\frac{\delta\mathcal{L}_n^{(l)}}{\|\mathbf{v}_{l,n}\|^2}\,\mathbf{v}_{l,n}\,\mathbf{h}_{l-1,n}^{\top}, \tag{8}$$

with N_l the number of perturbed units of layer l. The 15 noisy passes give $\lfloor 15/L \rfloor$ draws per layer: 5 for the MLP and the LSTM ($L = 3$), 3 for the ConvNet ($L = 5$) and 1 for the Transformer ($L = 15$). For the LSTM gate layer, called once per time step, the norm in Eq. (8) is taken over the injected noise of all calls, whereas the reference code of Dalm et al. [2026] (feed-forward only) uses the measured activity difference; the two coincide for a layer called once.

DUST. DUST [Dahal et al., 2026] is ported from the authors' reference implementation. A clean pass caches every layer's input $\mathbf{h}$ and output $\mathbf{a}$ and the per-unit losses ℓ_u (a unit is a token on the language models and an example on the image tasks). A *direct* draw jitters the outputs of one group of sites at every unit, $\mathbf{a} \to \mathbf{a} + \sigma\boldsymbol{\xi}$, reruns the forward pass and rewards each unit by its loss reduction $r_u = \ell_u - \tilde{\ell}_u$, centred across the K draws of the group; the estimated error at the site's output is $\hat{\mathbf{g}}_{\mathbf{a}_u} = -\frac{1}{K\sigma}\sum_k r_u^{(k)}\boldsymbol{\xi}_u^{(k)}$ and the weight gradient is $\sum_u \hat{\mathbf{g}}_{\mathbf{a}_u}\mathbf{h}_u^{\top}$ over the number of units, the same local outer product as the node rules. The groups follow the reference allocation: per Transformer block the attention and MLP output projections together, the MLP input layer alone and the attention output before its projection alone, plus the token and (our model only) the learned position embedding; every hidden layer of the MLP, ConvNet and LSTM is its own group. The 15 draws are split in proportion to the reference shares, front-loaded towards the first block (Transformer 3, 3, 1 and 2, 2, 1 for the two blocks' groups, 2 and 1 for the embeddings; LSTM gates 12, embedding 3; MLP 8, 7; ConvNet 5, 5, 3, 2). Queries, keys and values take *local* draws instead: their block's attention output is recomputed from the cached clean q, k, v with one of them jittered, and each head's change Δo_s is scored against the attention output's estimated error, $c_s = \langle \hat{\mathbf{g}}_{o_s}, \Delta o_s \rangle$; queries keep their own token's score, keys and values the scores of later tokens decayed by $\gamma = 0.98$ per token. The readout is jittered on the cached logits with four times the direct population (60 draws), only the cross-entropy recomputed, its reward centred over the batch. Local and head draws are partial recomputations and, as in the reference, are not counted as passes; their cost is in the wall-clock budget (Table 22). Departures from the reference: Adam with the tied ε instead of SGD with momentum and per-site scales (0.2, 0.4, 0.05), no residual mixing scalars, value embeddings or attention gates (absent from our model), and a position embedding treated like the token embedding. On the MLP, ConvNet and LSTM the rule reduces to layer-wise node perturbation with unit-local, draw-centred rewards and the cached-logit readout.

Decorrelated ANP (DANP). Decorrelated ANP [Dalm et al., 2026] is ANP in which every layer input is first decorrelated by a square matrix $\mathbf{R}_l$ that is not a parameter of the model, $\hat{\mathbf{h}}_{l-1} = \mathbf{R}_l\mathbf{h}_{l-1}$, with $\mathbf{R}_l$ initialised to the identity and trained by the local rule of Ahmad et al. [2023],

$$\mathbf{R}_l \leftarrow \mathbf{R}_l - \eta_R\left(\hat{\mathbf{C}}_l - \operatorname{diag}\hat{\mathbf{C}}_l\right)\mathbf{R}_l, \qquad \hat{\mathbf{C}}_l = \tfrac{1}{M}\textstyle\sum_{m=1}^{M}\hat{\mathbf{h}}_{l-1,m}\hat{\mathbf{h}}_{l-1,m}^{\top}, \tag{9}$$

from the decorrelated inputs recorded on the clean pass (M rows: examples, or examples $\times$ positions / patches). The decorrelation rate is fixed at $\eta_R = 10^{-3}$, the value of Dalm et al. [2026] and the default of their code, and is not tied to ε; $\mathbf{R}_l$ is updated once per step after the Adam step. Embeddings, whose inputs are one-hot, are not decorrelated.

Forward-forward (FF). Forward-forward [Hinton, 2022] trains each block with a local objective and never propagates an error between blocks. With goodness $G = \mathrm{mean}_j\, h_{l,j}^2$ (the mean of squares and threshold $\theta_{\text{FF}} = 2$ of the reference re-implementation of Pezeshki [2023]; Hinton uses the sum of squares), the local loss on a positive/negative pair is $\ell_{\text{FF}} = \log(1 + e^{\theta_{\text{FF}} - G_{\text{pos}}}) + \log(1 + e^{G_{\text{neg}} - \theta_{\text{FF}}})$, and activities are ℓ_2-normalised before the next block. A block is one hidden layer of the MLP, one convolution–ReLU–pool stage or the dense layer of the ConvNet, the LSTM layer, and one pre-LN block of the Transformer; for text the goodness is computed per position. For images, positive data carry the correct label overlaid on the first 10 pixels and negative data a wrong label; for text, negatives have 30 % of their characters replaced at random. The positive, negative and (for images) neutral-label inputs are forwarded as one batch, i.e. three N-sized passes per step on the image tasks and two on the text tasks. Two inference rules are reported: Hinton's label accumulation (the label whose overlay yields the highest total goodness over all blocks but the first, Section 3.3 of Hinton [2022]; we also log the all-blocks variant) and a linear readout trained on the normalised, detached top-block activity of the neutral-label pass (Hinton's softmax reads all hidden layers but the first; reading only the top block keeps the readout inside the parameter budget). The LSTM block's output lies in $(-1, 1)$, so its goodness cannot exceed 1; that block uses $\theta_{\text{FF}} = 0.25$.

Identity-path credit (IPC). IPC comes from the Noise speedrun [Gülbahar et al., 2026], where it underlies every speedrun record from its seventh on; we port it from the record scripts onto our Transformer (`zoo/optimizers/ipc.py`). It needs a residual stream, so it runs on the Transformer only. One clean pass caches every layer's input and output; every learning signal is then a measured change of a loss, or of a local output scored against a measured credit, from partial recomputations on the cached activations. (i) *Top credit*: each token's credit at the last residual stream, $\mathbf{g}_u = \partial \ell_u / \partial \mathbf{x}_{L,u}$, is measured by central differences along a full orthonormal basis $\{\mathbf{q}_j\}_{j=1}^{d}$ (a new random basis every step), each a re-evaluation of the final layer normalization, the readout and the token's cross-entropy at $\mathbf{x}_{L,u} \pm \varepsilon \mathbf{q}_j$: $\hat{\mathbf{g}}_u = \sum_j \frac{\ell_u(\mathbf{x}_{L,u} + \varepsilon \mathbf{q}_j) - \ell_u(\mathbf{x}_{L,u} - \varepsilon \mathbf{q}_j)}{2\varepsilon} \mathbf{q}_j$; the readout is a per-token function, so every token is probed at once without interference, and a complete basis returns the credit exactly up to the finite-difference error. (ii) *Identity path*: every earlier residual stream receives the same credit, so the attention and MLP output projections and the token and position embeddings take $\hat{\mathbf{g}}_u$ as their output error (the reference scales it by its residual mixing scalars, which our model lacks). The Jacobians of the later sub-blocks are dropped; this is the method's one approximation. (iii) *One local step*: each MLP hidden unit's pre-activation is moved along a full orthonormal cycle over the hidden width and the change of the block output, through the ReLU and the output layer, is scored against $\hat{\mathbf{g}}_u$; the target at the attention output before its projection is $\hat{\mathbf{g}}_u$ seen through that (linear) projection; values take their exact credit $\sum_s A_{st}\, \mathbf{t}_s$ (attention is linear in its values), queries a full orthonormal cycle per head with the attention rerun, and keys single-position, single-coordinate moves evaluated in closed form (one attention logit per query changes), each query's output change scored against its target $\mathbf{t}_s$. (iv) The readout takes the cross-entropy's central difference along every logit, in closed form on the cached logits. Weight gradients are output errors times cached inputs, as for the node rules. Every finite-difference step equals the tied ε, and the update is the shared Adam rule (the reference uses ANVIL II on the block matrices and steps relative to the probed vector's RMS). On our Transformer the estimate equals the true gradient for the readout and the top block's MLP (cosine > 0.9999); a unit test checks it against the identity-path rule computed by automatic differentiation (cosine > 0.9995 on every layer). IPC spends one forward pass per step; its partial recomputations are charged to the wall-clock budget.

Backpropagation (reference). One forward and one backward pass produce the exact gradient, fed to the same Adam update. It is the reference, not a zero-order method.

B Protocol details

Table 3 lists the four tasks of Section 4; the reporting conventions, the divergence rule, the ablations and the hardware and software of the protocol are specified in full below.

Reporting conventions. The primary metric is the *best validation checkpoint*: for each pair and seed, the evaluation with the lowest validation loss over the hour, whose test loss, test accuracy (classification) or bits per character (loss/ $\ln 2$, language modelling) is reported as the mean over seeds with the standard deviation in units of the last digit ("0.140(5)" is 0.140 ± 0.005; the sample

Table 2: Forward-pass budget per step. L is the number of perturbable layers (3 for the MLP and the LSTM, 5 for the ConvNet, 15 for the Transformer).

Method	Clean	Probe passes	Backward	Estimate per step
BP	1	0	full	exact gradient
SPSA	0	16	–	8 central differences, averaged
1.5-SPSA	1	16	–	8 curvature-scaled central differences
RS	1	15	–	best of 15 candidates
NP, ANP, DANP	1	15 (all layers)	–	15 noise draws, averaged (DANP: $+\mathbf{R}_t$ update)
1.5-ANP	1	16 (all layers, 8 pairs)	–	8 curvature-scaled central differences per unit
1.5-INP	1	16 (one layer each, 8 pairs)	–	as 1.5-ANP, attributed to the perturbed layer
INP	1	15 (one layer each)	–	$\lfloor 15/L \rfloor$ draws per layer
DUST	1	15 (one group each)	–	per-unit centred rewards; 60 head and local draws on cached outputs
FF	3 / 2	0	per block	local goodness gradients
IPC	1	0	–	exact top credit, residual path (Transformer only)

Table 3: Tasks, splits (train / validation / test) and architectures. Hidden nonlinearities are ReLU; convolutions are 3×3 with 2×2 max-pooling; the LSTM [Hochreiter and Schmidhuber, 1997] has one layer of 80 units; the Transformer [Vaswani et al., 2017] has two pre-LN blocks of width 48, 4 heads, feed-forward width 64, learned positional embeddings and context $T = 64$. *Units* is the number of pre-activation values one node-perturbation probe perturbs for a single example, summed over layers, time steps and spatial positions (the Transformer's position embedding, 64×48 values, is perturbed once per batch and counted once); it is the normalisation constant c of Eq. (6).

Key	Data	Split	Model	Params	Units
`mlp_mnist`	MNIST	55k / 5k / 10k	MLP 784–48–32–10	39,578	90
`conv_cifar10`	CIFAR-10	45k / 5k / 10k	conv16–conv32–conv32–fc48–10	39,450	26,682
`lstm_ptb`	PTB (char)	standard	emb32–LSTM80–V	41,810	25,728
`transformer_ptb`	PTB (char)	standard	emb48 + pos64–2 blocks–V	39,250	48,256

standard deviation over three seeds; cells without parentheses are single runs). The value at the end of the hour is reported as a secondary stability metric, and for the two classification tasks the checkpoint with the highest validation *accuracy* is reported as well (Appendix F), since loss and accuracy drift apart under a fixed step. FF on the image tasks is selected and reported by label-accumulation accuracy, with the readout accuracy in brackets. Mean ranks are over the ten probe methods, with a 95 % bootstrap interval over seeds. Test data are never used for selection.

Selection with disagreeing criteria. Where the stage-1 selection by best and by final validation loss disagree in a non-degenerate way (DANP on both image tasks), both ε are replicated and the better mean validation loss decides.

Divergence rule. Stage 2 stops a run and marks it diverged when a loss is non-finite or when its clean validation loss exceeds ten times its initial value at two consecutive evaluations. Stage 1 used a factor two and additionally compared each method's own training loss with its first-step value; the former stopped backpropagation on CIFAR-10 at 67 % accuracy and the latter was method-dependent because SPSA's training loss is a probe mean, so both were replaced; a run that the old rule did not stop is identical under the new one.

Ablations. (i) *Untied probe scale and step size*: for every probe method and task the grid $\eta \in \{10^{-2}, 10^{-3}, 10^{-4}\} \times \sigma \in \{10^{-2}, 10^{-3}, 10^{-4}\}$ plus $\sigma = 0.1$ at $\eta = 10^{-4}$, screened for 30 minutes at 1-minute granularity (the tied diagonal is taken from the one-hour runs at minute 30); the best untied setting of a pair is then replicated for one hour with three seeds where it beats the best tied diagonal cell by at least 3 % (at most twelve pairs, largest gains first; Table 9 reports the replications that completed before the deadline), so that a rule is also compared at its own best (η, σ) at equal compute and without any schedule. (ii) *The 1.5-order rules under their own recipe*: plain SGD instead of Adam, fp64 loss evaluation, 1.5-SPSA with $\eta = \sigma \in \{10^{-2}, 3 \cdot 10^{-3}, 10^{-3}, 3 \cdot 10^{-4}, 10^{-4}\}$ and 8 or 64 pairs with the $\alpha = 0$ (1SPSA) control, 1.5-ANP with $\eta \in \{0.05, 0.02, 0.01, 0.005\}$ decoupled from $\sigma \in \{10^{-2}, 10^{-3}\}$ with the $\alpha = 0$ control, a $\lambda = 10^{-2}$ floor (never selected) and, on the LSTM, the recurrent-state damping; 30-minute screening, single seed (Section 5.4). (iii) *Controls under the protocol*: 1.5-ANP with Adam at the step size the screening selected on the image tasks

($\eta = 10^{-4}$), the probe scale decoupled ($\sigma \in \{10^{-2}, 10^{-3}\}$), fp64 losses, the curvature exponent $\alpha \in \{0, 0.25, 0.5, 1\}$ and, on the LSTM, the recurrent-state damping (Table 8); 1.5-ANP with $\alpha = 0$ and with fp64 losses at $\varepsilon = 10^{-3}$; 1.5-SPSA with a non-inert factor ($\alpha = 0.5$, $\lambda = 10^{-3}$). (iv) *Estimator alignment*: the cosine between every estimator and the true gradient of a fixed batch as a function of the number of forward passes, per layer after a 2000-step backpropagation warm-up, in fp32, TF32 and fp64, for σ from 10^{-1} to 10^{-5} (Appendix F). (v) *Resources*: every run records its steps, seconds of training time per step and peak GPU memory (Appendix F); throughput is measured with eight runs sharing one host.

Not run. Two planned ablations did not run before the deadline and are reported nowhere: the isolated 150-second reruns of every selected configuration (throughput and peak memory are instead taken from the stage-2 runs, eight per host) and the $\eta = 10^{-3}$ off-diagonal cells of the untied grid for the six original probe methods; their job lists are in the repository.

Hardware and software. All runs use one RTX 5090 per run on an 8-GPU vast.ai node (8 runs at a time; PyTorch 2.9.1, CUDA 12.8), with the code at `https://github.com/Fchaubard/zero-order-ablations` (stage 1: commits `db100bc` and `f5424e9`; stage 2: `97c03f5`). Every run is logged to Weights & Biases (project `zoo-survey`) and writes a JSON file with its configuration, evaluation history and final metrics.

C Additional results and discussion

Table 4: The 1.5-order rules under their own recipe (plain SGD, fp64 losses, 30-minute screening): the best configuration by validation loss and its best-checkpoint test loss (accuracy or bits per character), the $\alpha = 0$ control at the same grid, and the protocol run (Adam, tied ε) at minute 30. Config key: lr $= \eta$; K17 / K129 = 8 / 64 antithetic pairs per step; sig $= \sigma$; lam $= \lambda$; l64 = fp64 losses.

Task	Rule	best config	test loss (acc / bpc)	$\alpha = 0$ control	protocol (Adam) @30 min
MNIST / MLP	1.5-SPSA	lr0.003 K129 l64 sgd	0.176 (94.8)	0.162 (94.9)	0.160±0.016
	1.5-ANP	lr0.01 sig0.01 lam0.001 l64 sgd	0.137 (96.0)	0.121 (96.6)	0.167±0.015
CIFAR-10 / ConvNet	1.5-SPSA	lr0.003 K129 l64 sgd	1.336 (51.7)	1.344 (51.6)	1.313±0.050
	1.5-ANP	lr0.005 sig0.001 lam0.001 l64 sgd	2.356 (20.5)	2.406 (9.4)	1.634±0.067
PTB / char-LSTM	1.5-SPSA	lr0.01 K17 l64 sgd	2.955 (4.26)	2.845 (4.10)	2.234±0.001
	1.5-ANP	lr0.05 sig0.01 lam0.0001 l64 sgd	2.463 (3.55)	2.424 (3.50)	2.167±0.021
PTB / char-Transformer	1.5-SPSA	lr0.01 K17 l64 sgd	2.258 (3.26)	2.259 (3.26)	2.230±0.004
	1.5-ANP	lr0.005 sig0.01 lam0.001 l64 sgd	2.763 (3.99)	2.763 (3.99)	2.585±0.076

Equal-pass view. Figure 6 shows the selected runs of Figure 3 against the number of forward passes (best validation checkpoint within the pass budget, mean over seeds), which removes the implementation- and host-dependent throughput of each method from the comparison. Backpropagation reaches 0.139 on MNIST and 0.907 on CIFAR-10 within 10^5 passes, where every probe rule is still near its initial loss except DANP (1.17 on CIFAR-10); at 10^6 passes NP and INP are at 0.107–0.108 on MNIST, within 0.006 of BP's best, and DANP and ANP at 1.09 and 1.16 on CIFAR-10 against 0.907; at 10^7 passes the image-task ordering is that of Table 1. On the LSTM the gap is larger at every budget: after 10^5 passes BP is at 1.24 against 2.17–2.28 for 1.5-ANP, INP and NP, because a backward pass through 64 time steps buys far more than a perturbed forward pass does.

Backpropagation reference. With Adam and no schedule, backpropagation completes 2.7–4.3M steps on MNIST (seed spread) and 2.0M on CIFAR-10 in the hour (Table 22) and overfits the 40k-parameter models within minutes. At $\eta = 10^{-3}$ the MNIST validation loss is lowest at the first evaluation (97.2 % test accuracy, three seeds) and the run ends at 97.4 % with a test loss of 2.4–2.6; the CIFAR-10 run has a validation loss above its initial value from the first evaluation on (its "best checkpoint" is step 0) and is stopped by the divergence rule at 66.9 %. At $\eta = 10^{-4}$ the CIFAR-10 run peaks at 69.2(6) % at minute 1 and ends at 65.6 % with a test loss of 14, and on MNIST $\eta = 10^{-5}$ gives the lowest validation loss (0.102(4) at minute 5, 97.0(2) %, three seeds), which the selection rule therefore picks; the three-seed $\eta = 10^{-4}$ run is equivalent in accuracy (97.2(1) %). The best point of BP on these models thus lies within the first few minutes at any learning rate, and the reference

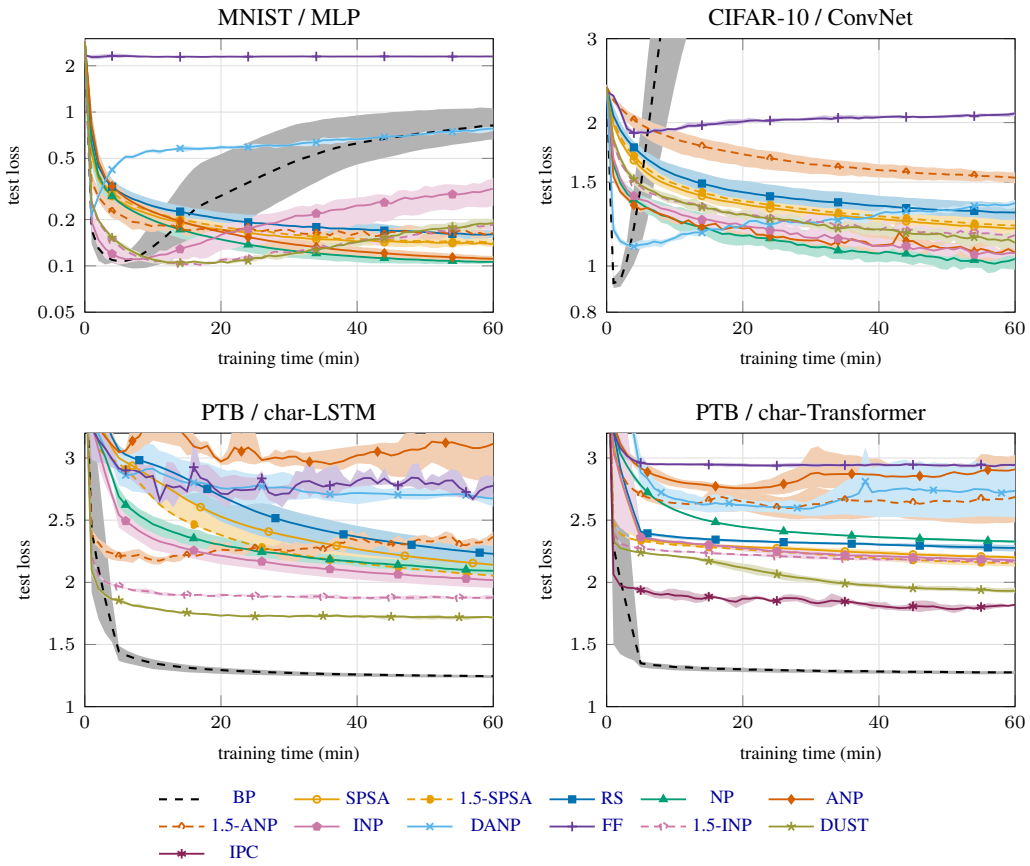


Figure 3: Test loss versus training time (minutes; evaluation excluded) of the selected 40k runs: seed mean (line) and seed range (shaded). Log y-axis on the image tasks, linear (clipped at 3.2) on the language models. The 1.5-order rules are dashed in their parent's colour; IPC runs on the Transformer only.

reported in Table 1 is the one the protocol's own rule selects. On the sequence tasks $\eta = 10^{-3}$ is best, BP is still improving at minute 60 (1.80 and 1.84 bits per character) and the reference is unproblematic.

Probe scale and step size. The stage-1 sweep (Appendix E, Figure 9) is decisive about the tied scalar: at $\varepsilon = 0.1$, 15 of the 24 runs of the six original probe methods diverge (all four DANP runs, three each of NP, ANP and RS, two of SPSA, none of INP), BP itself collapses on both image tasks at this step size, and the surviving probe runs make little progress; at $\varepsilon = 0.01$ seven probe runs diverge, five of them node-perturbation runs, and BP is stopped on MNIST; at $\varepsilon = 0.001$ no probe run diverges and 25 of the 32 original pairs select this value. The screening of the two 1.5-order rules under the stage-2 settings follows the same pattern: six of their sixteen runs at $\varepsilon \geq 0.01$ diverge (1.5-SPSA on both image tasks at 0.1 and 0.01, 1.5-ANP on CIFAR-10 and the LSTM at 0.1), and under the stage-2 settings every pair of the two 1.5-order rules then selected $\varepsilon = 0.001$ except 1.5-ANP on the LSTM (0.01), before the lower values were added (Table 16). Because ε is both the Adam step and the probe scale, these counts do not say whether a method diverged from noise or from step size, and the published rules do not obey the tie (Dalm et al. [2026] fix $\sigma = 10^{-3}$ with learning rates down to 10^{-5}; MeZO uses $\sigma = 10^{-3}$ with learning rates of 10^{-6}–10^{-7}). Table 6 separates the two, and Table 7 lists the controls under the protocol at $\varepsilon = 10^{-3}$: 1.5-SPSA with a non-inert factor ($\alpha = 0.5$, $\lambda = 10^{-3}$) is within 0.01 of SPSA on the image tasks and 0.02–0.11 worse on the sequence tasks, and 1.5-ANP with $\alpha = 0$ beats 1.5-ANP by 0.08 on MNIST and 0.58 on CIFAR-10 already at this ε, with fp64 losses alone recovering about half of that on MNIST and three quarters on CIFAR-10.

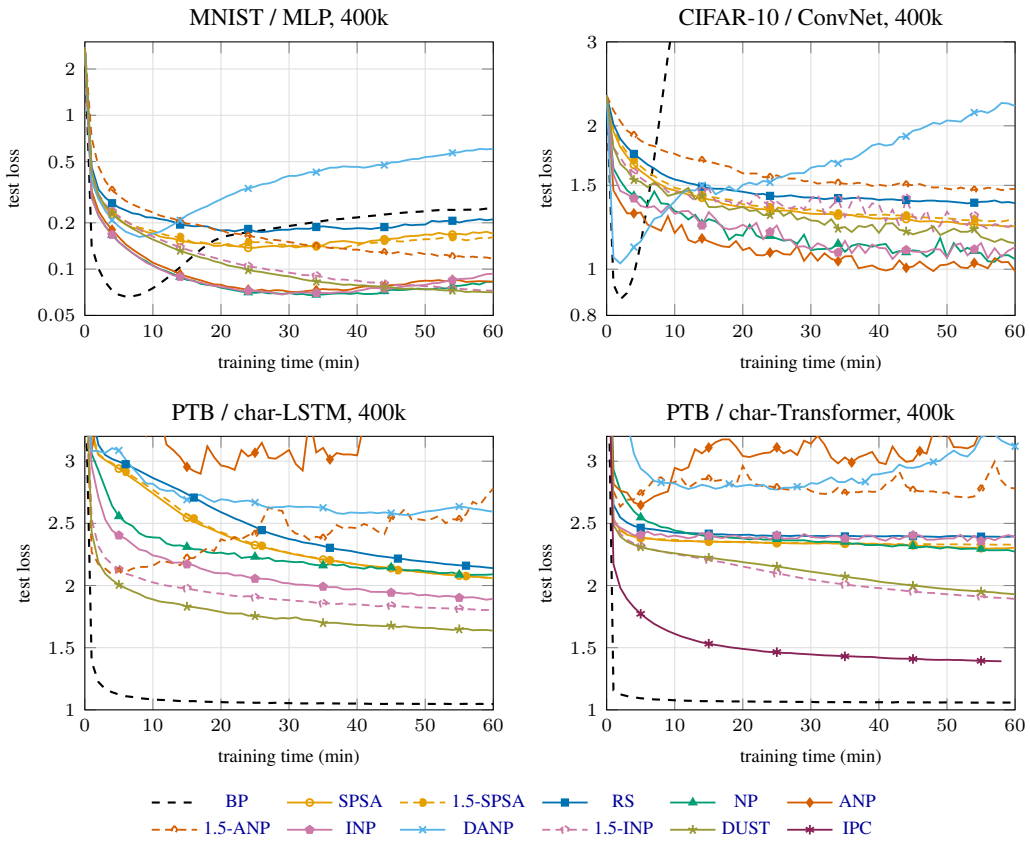


Figure 4: Test loss versus training time at 400k parameters (seed 0, ε selected at that size; Section 5.5); axes as in Figure 3. FF is omitted: its plotted quantity would be the readout cross-entropy, which is not what it trains.

Table 5: Backpropagation reference under the stage-2 protocol: test accuracy (%) or bits per character at the best validation checkpoint (with its minute) and at the end of the hour, for learning rates η and decoupled weight decay (AdamW). The reference of Table 1 is the row with the lowest validation loss on each task (10^{-5} on MNIST, 10^{-4} on CIFAR-10, 10^{-3} on both sequence tasks); the $\eta = 0.01$ image rows are stage-1 runs. A dagger marks a run stopped by the divergence rule (over-confident, not collapsed, in every daggered case).

		MNIST		CIFAR-10		LSTM		Transformer	
lr	decay	best (min)	final	best (min)	final	best (min)	final	best (min)	final
0.01	0	12.7 (0)	67.1†	62.7 (1)	59.3	1.85 (20)	1.93	1.89 (40)	1.90
0.001	0	97.2±0.2 (1)	97.4±0.0	9.4 (0)	66.9†	1.80±0.02 (60)	1.80±0.02	1.84±0.02 (57)	1.84±0.02
0.0001	0	97.2±0.1 (1)	97.1±0.0	69.2±0.6 (1)	65.6±0.3	2.12 (60)	2.12	1.93 (60)	1.93
0.00001	0	97.0±0.2 (5)	96.7±0.2	68.9 (9)	65.6	–	–	–	–
0.001	0.01	97.5 (4)	97.9	9.4 (0)	67.4	1.78 (60)	1.78	1.86 (60)	1.86
0.0001	0.01	97.3 (1)	97.3	69.0 (1)	65.9	1.94 (60)	1.94	1.89 (59)	1.89

Estimator alignment. Figure 8 measures, on one fixed batch per task, the cosine between each estimator and the true gradient as a function of the number of forward passes spent on it, after a 2000-step backpropagation warm-up, in fp32 on the GPU with TF32 disabled (Appendix F gives the per-layer values and the dependence on the probe scale). On the MLP the node-perturbation rules obtain, from the protocol's 16 passes, a direction with cosine 0.4–0.6 to the gradient, rising to 0.96 (NP), 0.95 (INP), 0.83 (DANP) and 0.81 (ANP) with 1024 passes, whereas the weight-space rules get 0.01 from 16 passes and 0.07–0.11 from 1024, the $\sqrt{K/P}$ scaling of a 40k-dimensional random search. On the ConvNet every estimator is below 0.1 at 16 passes and the node rules reach 0.51–0.53 with 1024: the convolutional layers, whose 26,682 perturbed unit-positions approach

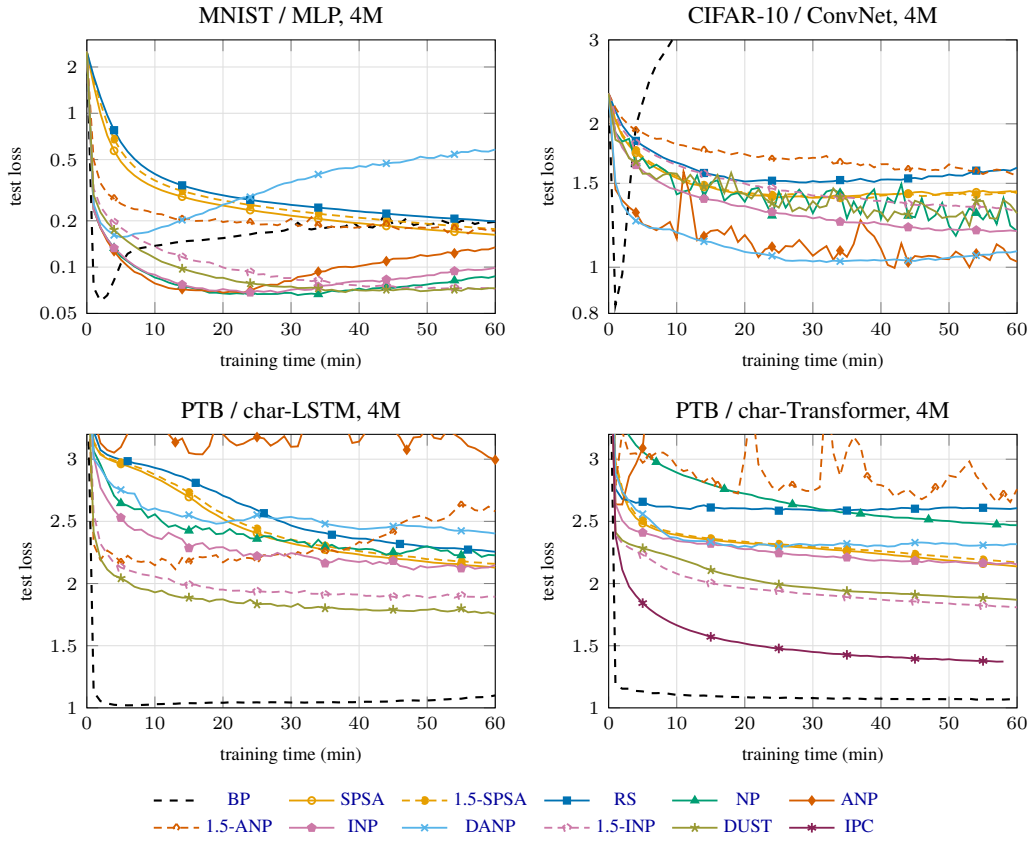


Figure 5: Test loss versus training time at 4M parameters, as in Figure 4.

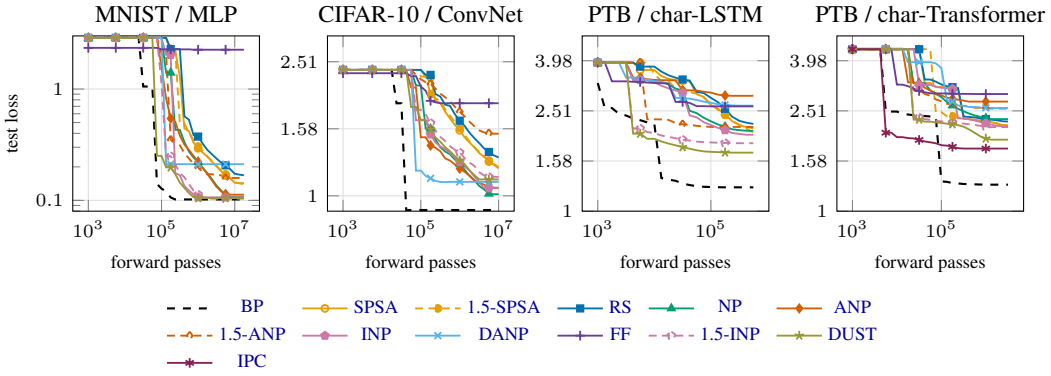


Figure 6: Equal-compute view: test loss versus the number of forward passes (one BP step counts one pass plus its backward), interpolated between the 1-minute evaluations, mean over seeds.

the number of weights, need thousands of probes before the node estimate carries signal (Table 26: 0.4–0.6 in the three convolutional layers against 0.8–1.0 in the two dense layers), which is why the node rules train the ConvNet slowly and the weight-space rules hardly at all. On the LSTM and the Transformer the whole-model cosine stays below 0.26 and 0.17 at 1024 passes (INP reaches 0.40 and 0.33 at 4096): perturbations propagate through shared weights over 64 positions and decorrelate the loss difference from any single unit. The curvature preconditioners behave as their results suggest: 1.5-SPSA is indistinguishable from SPSA at every probe count and every σ, because with $\alpha = 0.1$ and $\lambda = 1$ its per-direction weights span a narrow range whose common scale Adam removes, while 1.5-ANP's per-unit factor ($\alpha = 0.5$) lowers the alignment of ANP from 0.81 to 0.69 on the MLP and

Table 6: Probe scale decoupled from the step size: test loss of the best validation checkpoint within 30 minutes for every probe method and task. Off-diagonal cells are single 30-minute runs; the tied diagonal cells are one-hour runs at minute 30 (the stage-2 seeds where that value was selected, the stage-1 run otherwise). The $\eta = 10^{-3}$ off-diagonal cells and the ($\eta = 10^{-2}$, $\sigma = 10^{-3}$) cell were run only for the rules that show them; the last column is the $\sigma = 0.1$ row added at $\eta = 10^{-4}$. A dagger marks a diverged run.

		$\eta = 10^{-2}$			$\eta = 10^{-3}$			$\eta = 10^{-4}$			
Task	Method	σ=10^{-2}	10^{-3}	10^{-4}	σ=10^{-2}	10^{-3}	10^{-4}	σ=10^{-1}	10^{-2}	10^{-3}	10^{-4}
MNIST / MLP	SPSA	2.229	–	2.001	–	0.365	–	0.209	0.165	0.173	**0.154**
	1.5-SPSA	2.226†	2.159	1.977	0.332	0.365	0.392	0.222	0.171	0.180	**0.160**
	RS	2.761	2.496	2.308	–	0.440	–	2.767	0.206	0.208	**0.182**
	NP	0.984	–	0.503	–	0.143	–	**0.106**	0.110	0.112	0.107
	ANP	0.557	–	0.496	–	0.126	–	0.110	0.109	**0.107**	0.112
	1.5-ANP	1.574	1.987	2.101	0.154	0.218	0.324	0.142	**0.127**	0.146	0.167
	INP	0.606†	–	0.370	–	0.134	–	0.111	0.110	0.115	**0.107**
	DANP	0.306	0.299	0.303	–	**0.211**	–	0.227	0.216	0.215	0.215
CIFAR-10 / ConvNet	SPSA	3.234†	–	2.303	–	1.645	–	2.085	1.304	1.353	**1.292**
	1.5-SPSA	2.367†	5.402	2.993	1.525	1.609	1.604	2.106	1.322	1.373	**1.313**
	RS	2.328†	2.406†	9.260†	–	1.730	–	2.406	1.463	1.447	**1.373**
	NP	2.406†	–	2.331†	–	1.407	–	1.227	1.192	1.188	**1.105**
	ANP	2.312	–	2.306	–	1.175	–	1.710	1.262	1.161	**1.122**
	1.5-ANP	2.304	2.304	2.306	**1.360**	1.775	1.860	1.508	1.428	1.473	1.639
	INP	2.304	–	2.303	–	1.315	–	1.597	1.159	1.165	**1.139**
	DANP	1.144	1.170	1.174	–	**1.101**	–	2.081†	1.444	1.258	1.238
PTB / char-LSTM	SPSA	**2.109**	–	2.129	–	2.370	–	3.192	3.100	3.090	3.086
	1.5-SPSA	**2.095**	2.150	2.127	2.217	2.234	2.224	3.231	3.102	3.114	3.107
	RS	2.306	**2.250**	2.264	–	2.483	–	3.508	3.253	3.270	3.272
	NP	2.218	–	**2.205**	–	2.222	–	2.971	2.949	2.946	2.987
	ANP	3.356†	–	2.999	–	**2.961**	–	3.416	3.008	3.007	3.008
	1.5-ANP	2.167	**2.139**	2.139	2.236	2.268	2.235	2.616	2.628	2.718	2.693
	INP	**2.023**	–	2.033	–	2.137	–	2.839	2.837	2.826	2.816
	DANP	3.132†	2.758	2.857	–	**2.730**	–	3.633	3.059	3.051	3.059
PTB / char-Transformer	SPSA	2.536	–	2.580	–	**2.263**	–	2.892	2.348	2.352	2.352
	1.5-SPSA	2.547	2.584	2.570	**2.209**	2.230	2.225	2.920	2.358	2.360	2.359
	RS	2.736	2.792	2.768	–	**2.317**	–	3.551	2.721	2.401	2.400
	NP	2.964	–	2.937	–	**2.372**	–	2.477	2.397	2.399	2.393
	ANP	4.496†	–	2.827	–	3.065	–	3.536†	2.936	**2.719**	2.735
	1.5-ANP	2.703	2.651	2.694	**2.581**	2.593	2.633	2.595	2.675	2.671	2.585
	INP	2.605	–	2.660	–	**2.228**	–	2.345	2.329	2.324	2.332
	DANP	2.588	2.414	**2.384**	–	2.525	–	4.235	3.770	3.066	3.101

Table 7: Controls under the protocol at $\varepsilon = 10^{-3}$ (Adam, tied scale): test loss (accuracy or bits per character) at the best validation checkpoint of the tied runs of SPSA, 1.5-SPSA and 1.5-ANP (mean over seeds) next to single-seed controls: 1.5-SPSA with a non-inert curvature factor ($\alpha = 0.5$, $\lambda = 10^{-3}$), 1.5-ANP without curvature weighting ($\alpha = 0$) and 1.5-ANP with fp64 losses. The LSTM selection of 1.5-ANP is $\varepsilon = 10^{-2}$, so its 10^{-3} run is listed; its two LSTM controls were not run, and SPSA on CIFAR-10 at 10^{-3} is a single stage-1 run.

Task	SPSA	1.5-SPSA	1.5-SPSA α=0.5, λ=10^{-3}	1.5-ANP	1.5-ANP α=0	1.5-ANP fp64
MNIST / MLP	0.365±0.001	0.365±0.013	0.364 (90.0)	0.224±0.023	0.145 (96.3)	0.187 (95.2)
CIFAR-10 / ConvNet	1.645	1.609±0.011	1.638 (38.7)	1.775±0.018	1.200 (58.1)	1.349 (52.9)
PTB / char-LSTM	2.141±0.082	2.054±0.008	2.249 (3.24)	2.268	–	–
PTB / char-Transformer	2.200±0.017	2.155±0.026	2.216 (3.20)	2.593±0.071	2.536 (3.66)	2.615 (3.77)

from 0.52 to 0.27 on the ConvNet at $\sigma = 10^{-3}$ and raises it on the LSTM (0.26 against 0.04); its second difference is the only quantity that degrades with the probe scale in fp32 (Table 24: 0.74 at $\sigma = 10^{-2}$ to 0.46 at 10^{-5} on the MLP), the first-difference estimators being unchanged down to 10^{-5}. Numerical precision also matters for the convolutions: with TF32, which PyTorch enables by default, the ConvNet cosine of NP at $\sigma = 10^{-3}$ and 1024 passes drops from 0.445 to 0.420 and that of INP from 0.425 to 0.358 (fp64: 0.475 and 0.484), while SPSA and ANP are unaffected and nothing changes at $\sigma = 10^{-2}$ (Table 27, measured on a different batch and warm-up seed from Table 26). Stage 1 ran with TF32 on; stage 2 runs in full fp32.

Table 8: 1.5-ANP under the protocol's optimizer at the step size the ε screening selected on the image tasks (Adam, $\eta = 10^{-4}$), with the probe scale decoupled from the step size and the losses evaluated in fp64: test loss (accuracy or bits per character) at the best validation checkpoint, seed 0, for $\sigma \in \{10^{-2}, 10^{-3}\}$ and curvature exponent $\alpha \in \{0.5, 0, 0.25, 1\}$ ($\alpha = 0$ removes the curvature weighting); the "damped" row is the LSTM with the recurrent-state damping (0.5) of its specification. The last two columns are the tied-protocol selections of 1.5-ANP and ANP (mean over seeds; the CIFAR-10 selection of 1.5-ANP is $\varepsilon = 10^{-5}$, its 10^{-4} run is within 0.05 of it).

Task	$\sigma{=}10^{-2}$	10^{-2}, $\alpha{=}0$	10^{-3}	10^{-3}, $\alpha{=}0$	10^{-3}, $\alpha{=}\frac{1}{4}$	10^{-3}, $\alpha{=}1$	1.5-ANP tied	ANP tied
MNIST / MLP	0.132 (96.2)	0.115 (96.8)	0.133 (96.2)	0.112 (97.0)	0.119 (96.5)	0.204 (94.4)	0.159±0.007	0.112±0.003
CIFAR-10 / ConvNet	1.279 (56.9)	1.222 (56.5)	1.307 (54.7)	1.120 (60.4)	1.174 (58.8)	1.981 (31.2)	1.532±0.041	1.058±0.025
PTB / char-LSTM	2.482 (3.58)	2.475 (3.57)	2.537 (3.66)	2.475 (3.57)	2.395 (3.46)	2.848 (4.11)	2.167±0.021	2.885±0.082
PTB / char-LSTM, damped	2.382 (3.44)	–	2.383 (3.44)	–	–	–	–	–
PTB / char-Transformer	2.670 (3.85)	2.552 (3.68)	2.664 (3.84)	2.560 (3.69)	2.587 (3.73)	2.641 (3.81)	2.566±0.085	2.735±0.039

Table 9: Untied settings replicated for an hour: for the pairs whose best untied (η, σ) beat the best tied cell of Table 6 by at least 3 %, the one-hour best-checkpoint test loss at that setting (mean over the seeds that completed before the deadline) next to the tied selection of Table 1.

Task	Method	untied (η, σ)	untied (seeds)	tied selection (seeds)	ε^*
MNIST / MLP	ANP	(0.0001, 0.001)	0.110±0.005 (3)	0.112±0.003 (3)	1e-05
MNIST / MLP	1.5-ANP	(0.0001, 0.01)	0.127±0.004 (3)	0.159±0.007 (3)	0.0001
CIFAR-10 / ConvNet	1.5-ANP	(0.001, 0.01)	1.349±0.064 (3)	1.532±0.041 (3)	1e-05
PTB / char-Transformer	DANP	(0.01, 0.0001)	2.389±0.029 (3)	2.578±0.092 (2)	0.01

Sequence models. On the test split a uniform model scores 5.64 bits/char, a unigram 4.32, a bigram 3.31 and an add-0.1 trigram 2.61 (count models estimated on the training text; Appendix G). On the LSTM, INP, 1.5-SPSA, NP, SPSA, 1.5-ANP and RS (2.91–3.22) sit between the bigram and the trigram; FF (3.78) and DANP (3.80) are worse than the bigram and ANP (4.16) is near the unigram. On the Transformer only 1.5-SPSA, INP, SPSA and RS (3.11–3.28) edge past the bigram. BP reaches 1.80 and 1.84 bits per character, well beyond the trigram: after one hour the zero-order rules have learned roughly bigram statistics and backpropagation a real character model. The curves of the leading probe rules are still descending at minute 60, so part of the gap is budget rather than estimator, but at 12–34k steps per hour on the LSTM closing it would take days.

Forward-forward. FF is not a probe method and trains fast (1.7M steps per hour on the MLP), but its two inference rules disagree. The linear readout on the neutral-label features of the 32-unit top block peaks within the first minute on MNIST (87.5 % at $\varepsilon = 0.001$) and then collapses to 11–14 % as its training cross-entropy rises towards $\ln 10$; Hinton's label accumulation on the same trained blocks, summed over all blocks but the first (the protocol's rule, which on this two-block MLP means the second block alone), gives 95.2 % for the screening seed at the selected $\varepsilon = 0.1$ but 59.7(295) % over three seeds (the other two average 42 %) and 93.1(3) % at $\varepsilon = 0.01$ (Table 16). On CIFAR-10 both numbers are 27–34 %, and on the sequence tasks FF with corrupted-sequence negatives is third-worst on the LSTM and worst on the Transformer (3.78 and 4.23 bits per character): its goodness separates real from corrupted text, but the normalised features support only a weak next-character readout.

Faithfulness. SPSA and 1.5-SPSA use Gaussian rather than Rademacher directions and RS is expressed through Adam (Table 11: the direct-jump variant); 1.5-ANP applies the recurrent-state damping of its specification only in the controls of Table 8; FF's goodness, threshold and readout follow the reference re-implementation and the parameter budget rather than Hinton's 2,000-unit layers; DANP's decorrelation rate is fixed and un-normalised, which is why it reaches NaN within 21 steps at $\varepsilon = 0.1$ on both image models (a single Adam step inflates the ReLU activities by an order of magnitude and the $\mathbf{R}_l$ update then grows tenfold per step). All of these are stated where the method is defined.

D Extended related work

This appendix is the unabridged version of Section 2.

Table 10: Model-size sweep: best-checkpoint test loss per task and summed, one seed, at the ε with the lowest validation loss among those run at that size (the 40k rows are the seed means of Table 1). Widths: MLP hidden (400, 240) and (1600, 1700); ConvNet channels (48, 96, 96) and (128, 256, 256); LSTM hidden 265 and 870; Transformer width 152 and 496. IPC runs on the Transformer only; its 400k and 4M runs stopped at minute 58 of 60.

Params	Task	BP	SPSA	1.5-SPSA	RS	NP	ANP	1.5-ANP	INP	1.5-INP	DUST	DANP	FF	IPC
40k	MNIST / MLP	0.102	0.142	0.140	0.168	0.107	0.112	0.159	0.107	0.104	0.105	0.211	2.335	–
	CIFAR-10 / ConvNet	0.907	1.200	1.215	1.292	1.011	1.058	1.532	1.055	1.139	1.119	1.101	2.007	–
	PTB / char-LSTM	1.245	2.141	2.054	2.229	2.092	2.885	2.167	2.017	1.867	1.712	2.635	2.618	–
	PTB / char-Transformer	1.275	2.200	2.155	2.276	2.328	2.735	2.566	2.176	2.164	1.928	2.578	2.932	1.777
	sum	3.529	5.683	5.563	5.965	5.539	6.791	6.425	5.355	5.274	4.864	6.526	9.893	–
400k	MNIST / MLP	0.068	0.141	0.141	0.180	0.071	0.073	0.119	0.071	0.074	0.075	0.162	2.302	–
	CIFAR-10 / ConvNet	0.864	1.237	1.265	1.387	1.037	1.005	1.468	1.062	1.227	1.140	1.028	1.821	–
	PTB / char-LSTM	1.047	2.059	2.059	2.140	2.086	2.898	2.088	1.886	1.804	1.637	2.567	2.525	–
	PTB / char-Transformer	1.058	2.297	2.329	2.391	2.273	2.646	2.634	2.357	1.892	1.930	2.772	2.856	1.391
	sum	3.037	5.734	5.794	6.097	5.466	6.621	6.308	5.376	4.998	4.782	6.529	9.504	–
4M	MNIST / MLP	0.060	0.162	0.177	0.198	0.067	0.072	0.168	0.071	0.074	0.070	0.157	2.139	–
	CIFAR-10 / ConvNet	0.832	1.409	1.417	1.510	1.164	1.011	1.555	1.187	1.310	1.221	1.025	1.771	–
	PTB / char-LSTM	1.021	2.128	2.159	2.256	2.214	2.995	2.113	2.111	1.882	1.754	2.402	2.626	–
	PTB / char-Transformer	1.067	2.137	2.171	2.586	2.470	2.614	2.655	2.154	1.809	1.869	2.292	2.795	1.371
	sum	2.981	5.836	5.923	6.550	5.914	6.692	6.492	5.522	5.075	4.915	5.876	9.330	–

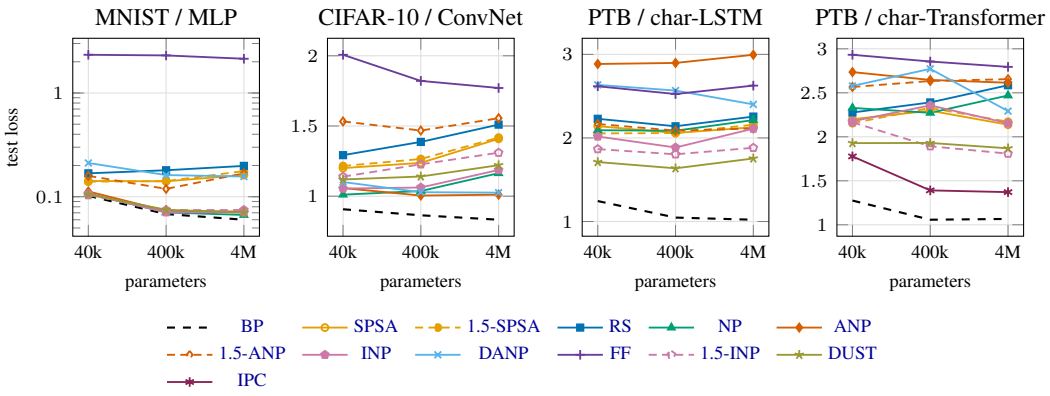


Figure 7: Per-task best-checkpoint test loss against model size (one seed at 400k and 4M, seed mean at 40k; ε re-selected per size; 1.5-INP and DUST run at the 40k ε and a decade below; IPC, Transformer only, at three ε per size, its 400k and 4M runs stopped at minute 58 of 60).

Weight-space perturbation. Simultaneous perturbation stochastic approximation [Spall, 1992, 1998] estimates a gradient from two loss evaluations at $\boldsymbol{\theta} \pm \sigma \mathbf{z}$ with a random direction $\mathbf{z}$; with Gaussian $\mathbf{z}$ the estimator is the gradient of the Gaussian-smoothed loss [Nesterov and Spokoiny, 2017], with complexity understood in the convex [Duchi et al., 2015] and bandit [Flaxman et al., 2005] settings (see Liu et al. [2020] for a survey). Weight perturbation was proposed for analog VLSI training by Jabri and Flower [1992], with stochastic error descent [Cauwenberghs, 1992] and summed-weight perturbation [Flower and Jabri, 1992] as early variance reductions. Evolution strategies use the same estimator with many directions, fitness shaping and natural-gradient corrections [Wierstra et al., 2014, Hansen et al., 2003, Hansen, 2016]; Salimans et al. [2017] and Mania et al. [2018] showed them competitive with policy-gradient reinforcement learning, Zhang et al. [2017] related them to SGD, and Lenc et al. [2019] applied them to supervised learning. Adaptive-moment updates on top of zero-order estimates were studied by Chen et al. [2019], and Silver et al. [2022] analysed learning from directional derivatives along random directions. MeZO [Malladi et al., 2023] fine-tunes billion-parameter language models with two-point SPSA and in-place perturbation; 1.5-SPSA [Chaubard, 2025b] adds a three-point curvature estimate along each direction to damp the step in sharp regions. DeepZero [Chen et al., 2024] and Zhang et al. [2024] benchmark zero-order estimators on large pretrained models; we vary the estimator *and* the architecture at a scale where every method can be run to its one-hour limit. Our SPSA, 1.5-SPSA and RS baselines are the two-point, curvature-preconditioned and best-of-K instances of this family.

Table 11: Follow-up runs (seed 0, 1 h). Top: random search through Adam (protocol) against the direct jump $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \sigma \mathbf{z}_{k^*}$ at the three model sizes.

Params	Task	RS (Adam, protocol)	RS direct jump
40k	MNIST / MLP	0.168±0.012 (94.9±0.3)	0.171 (94.7)
	CIFAR-10 / ConvNet	1.292±0.056 (53.7±2.2)	1.287 (54.3)
	PTB / char-LSTM	2.229±0.092 (3.22±0.13)	2.139 (3.09)
	PTB / char-Transformer	2.276±0.027 (3.28±0.04)	2.294 (3.31)
400k	MNIST / MLP	0.180 (94.4)	0.182 (94.3)
	CIFAR-10 / ConvNet	1.387 (50.9)	1.361 (51.7)
	PTB / char-LSTM	2.140 (3.09)	2.154 (3.11)
	PTB / char-Transformer	2.391 (3.45)	2.389 (3.45)
4M	MNIST / MLP	0.198 (94.4)	0.239 (92.8)
	CIFAR-10 / ConvNet	1.510 (46.7)	1.519 (45.5)
	PTB / char-LSTM	2.256 (3.25)	2.261 (3.26)
	PTB / char-Transformer	2.586 (3.73)	2.626 (3.79)

Table 12: Follow-up runs (seed 0, 1 h): 65 forward passes per step at Adam $\eta = 10^{-4}$, $\sigma = 10^{-3}$ (1.5-ANP with 32 antithetic pairs and fp64 losses, with its $\alpha = 0$ control; ANP and INP with 64 noisy passes) against the 17- and 16-pass tied-protocol selections of Table 1.

Task	1.5-ANP 65	1.5-ANP 65, α=0	ANP 65	INP 65	1.5-ANP 17	ANP 16	INP 16
MNIST / MLP	0.115 (96.8)	0.106 (96.8)	0.108 (96.6)	0.107 (96.8)	0.159±0.007	0.112±0.003	0.107±0.004
CIFAR-10 / ConvNet	1.282 (56.5)	1.119 (60.7)	1.059 (63.5)	1.032 (64.2)	1.532±0.041	1.058±0.025	1.055±0.015
PTB / char-LSTM	2.660 (3.84)	2.499 (3.61)	3.032 (4.37)	2.647 (3.82)	2.167±0.021	2.885±0.082	2.017±0.055
PTB / char-Transformer	2.621 (3.78)	2.507 (3.62)	2.664 (3.84)	2.263 (3.27)	2.566±0.085	2.735±0.039	2.176±0.022

Node perturbation. Node perturbation injects noise into the units of the network and correlates the resulting loss change with the noise, which yields a weight update local to each layer [Widrow and Lehr, 1990, Werfel et al., 2003, 2005, Fiete and Seung, 2006]. Hiratani et al. [2022] analysed its stability and showed that it suffers from weight diffusion in mismatched networks; Züge et al. [2023] compared weight and node perturbation on temporally extended tasks and found weight perturbation often similar or better, a finding our sequence tasks revisit. Dalm et al. [2026] reformulated node perturbation as a directional derivative (*iterative* NP), proposed *activity-based* NP that uses the measured activity change rather than the injected noise, and combined it with the input decorrelation rule of Ahmad et al. [2023], which also accelerates backpropagation [Dalm et al., 2024, Ahmad, 2024]. 1.5-ANP [Chaubard, 2025a] applies a three-point curvature preconditioner to central-difference activity perturbation. We implement all five variants.

Local rules and benchmarks. The forward-forward algorithm [Hinton, 2022] replaces the backward pass by two forward passes with positive and negative data and trains each layer to separate them with a local "goodness" objective. Rules that broadcast a global error vector [Clark et al., 2021], feedback alignment [Lillicrap et al., 2016, Nøkland, 2016] and synthetic gradients [Jaderberg et al., 2017] still require a learned or random backward signal, and forward gradients [Baydin et al., 2022, Ren et al., 2023] require a Jacobian-vector product through the network; they are outside the zero-order scope of this benchmark, which includes forward-forward as the representative local rule. Identity-path credit, from the Noise speedrun [Gülbahar et al., 2026] that followed DUST, is closest to the global-error rules: it broadcasts one error vector to every block, but it *measures* that vector with forward differences of the loss and uses the residual identity as its backward path, so it needs neither a backward signal nor a Jacobian-vector product; the speedrun's later records add exact forward differences through the later blocks with attention held fixed, cross-token probes and weight averaging, which we do not port. Controlled comparisons of biologically motivated rules exist for the feedback-alignment family [Bartunov et al., 2018, Launay et al., 2020, Sanfiz and Akrout, 2021]; we are not aware of one that puts weight perturbation, the node-perturbation family and forward-forward on the same models under one budget.

E Stage-1 screening grid

Table 17 lists the final validation and test loss of every stage-1 run of the eight original methods, Figure 9 the final test loss against ε, and Figure 10 every run's test-loss curve. The validation and test

Table 13: 1.5-INP: the 1.5-ANP estimator (antithetic pairs, per-unit normalisation, token-local attribution, fp64 losses) with INP's one-layer-at-a-time probing, eight pairs per step cycling over the layers: best-checkpoint test loss (accuracy or bits per character). $\alpha = 0.5$: one seed at INP's ε. $\alpha = 0$: mean $\pm$ s.d. over three seeds at the tied ε with the best seed-0 validation loss (Table 14). Untied: the (ε, σ) pair with the best seed-0 validation loss among $\sigma \in \{10\varepsilon, 100\varepsilon\}$ at the selected ε (only 10ε on the Transformer, with three seeds; one seed otherwise; not run on the LSTM; INP's untied grid, Table 6, has no run at these pairs). Last two columns: the tied selections of INP and 1.5-ANP.

Task	α=0.5	ε	α=0	(ε, σ)	α=0, untied	INP	1.5-ANP
MNIST / MLP	0.134 (96.2)	10^{-4}	0.104±0.002 (97.0±0.1)	$(10^{-4}, 10^{-2})$	0.101 (97.0)	0.107±0.004	0.159±0.007
CIFAR-10 / ConvNet	1.395 (50.2)	10^{-4}	1.139±0.015 (60.0±0.5)	$(10^{-4}, 10^{-3})$	1.176 (58.7)	1.055±0.015	1.532±0.041
PTB / char-LSTM	2.166 (3.12)	10^{-2}	1.867±0.013 (2.69±0.02)	–	–	2.017±0.055	2.167±0.021
PTB / char-Transformer	2.357 (3.40)	10^{-3}	2.164±0.010 (3.12±0.01)	$(10^{-3}, 10^{-2})$	2.146±0.020 (3.10±0.03)	2.176±0.022	2.566±0.085

Table 14: 1.5-INP step-size bracket ($\alpha = 0$, seed 0, tied $\sigma = \varepsilon$): best-checkpoint test loss at each ε; the ε selected by validation loss in bold; – not run.

Task	10^{-5}	10^{-4}	10^{-3}	10^{-2}	10^{-1}
MNIST / MLP	0.131	**0.103**	0.161	–	–
CIFAR-10 / ConvNet	1.369	**1.152**	1.467	–	–
PTB / char-LSTM	–	2.229	1.907	**1.857**	2.108
PTB / char-Transformer	–	2.222	**2.174**	2.309	–

values of a collapsed run can differ by orders of magnitude at the same evaluation because the loss of a collapsed network is dominated by a few examples with huge logits. The two 1.5-order rules and the values $\varepsilon \leq 10^{-4}$ of every method were screened under the stage-2 settings (1-minute evaluations, fp32, 1 h); Table 16 gives the validation loss of every value and the resulting selections.

F Selected configurations, throughput and ablation tables

Table 18 lists the selected ε and the minute of the best checkpoint of every pair, Table 20 the best-validation-*accuracy* checkpoints of the two classification tasks, Table 19 the end-of-hour values, Tables 21–23 the throughput, step counts and peak memory, and Tables 24–27 the alignment against the probe scale, the per-layer alignment and the precision check.

G Implementation details and hyper-parameters

Perturbable layers. Every trainable layer of every model is a perturbable linear, convolutional or embedding layer. A global context selects what a layer does on its forward pass: (i) a plain differentiable forward (backpropagation, forward-forward local losses, evaluation); (ii) a *clean* forward that detaches the layer input and records the input $\mathbf{h}_{l-1}$ and pre-activation $\mathbf{a}_l$, keeping a local autograd graph to the layer's own parameters only; or (iii) a graph-free *noisy* forward that adds $\pm\sigma\boldsymbol{\xi}_l$ to the pre-activation of every layer (or of one selected layer for INP; the minus sign reuses the probes of the plus pass for the antithetic pairs of 1.5-ANP) and records $\boldsymbol{\xi}_l$ and $\tilde{\mathbf{a}}_l$. Given a per-unit credit tensor $\mathbf{G}_l$, the weight update $\mathbf{G}_l^\top \mathbf{h}_{l-1}$ and its convolutional and embedding analogues are obtained by a one-layer backward through the recorded local graph, so no error is propagated through any nonlinearity. A layer called several times in one forward (the LSTM gate layer, once per time step) accumulates one record per call.

Decorrelation. Each linear and convolutional layer of DANP owns a square matrix $\mathbf{R}_l$ (identity-initialised buffer, not an optimizer parameter) that multiplies the layer input; for convolutions the decorrelated vector is the unfolded $C_{\text{in}}k^2$ patch and the decorrelation is itself a convolution. $\mathbf{R}_l$ is updated once per step with Eq. (9) from the inputs recorded on the clean pass, from the first step. Embeddings are not decorrelated.

1.5-ANP. Attribution units are examples (images) or (sequence, position) tokens (text). The per-token loss is the next-character cross-entropy at that position; the per-token norm $\|\mathbf{d}_u\|^2$ sums the squared activity differences of every layer at that position (token embedding; the gate pre-activations of the LSTM call at that time step; the projections and feed-forward layers of the Transformer at

Table 15: DANP with the decorrelation rate η_R lowered from the protocol's 10^{-3} (one seed, at the ε and size shown): best-checkpoint test loss (accuracy or bits per character), the minute it was reached and the end-of-hour loss. The $\eta_R = 10^{-3}$ rows are the protocol's own runs at the same setting (the LSTM and Transformer references are the seed means of Table 1).

Task	Params	ε	η_R	best	minute	final
MNIST / MLP	40k	10^{-3}	10^{-3}	0.213 (93.9)	1	0.792
MNIST / MLP	40k	10^{-3}	10^{-4}	0.158 (95.3)	1	1.141
MNIST / MLP	40k	10^{-3}	10^{-5}	0.132 (96.2)	1	1.030
MNIST / MLP	4M	10^{-4}	10^{-3}	0.157 (95.6)	5	0.582
MNIST / MLP	4M	10^{-4}	10^{-4}	0.120 (96.3)	2	0.503
CIFAR-10 / ConvNet	40k	10^{-3}	10^{-3}	1.094 (64.1)	8	1.378
CIFAR-10 / ConvNet	40k	10^{-3}	10^{-4}	1.090 (62.0)	6	1.401
CIFAR-10 / ConvNet	40k	10^{-3}	10^{-5}	1.114 (64.0)	14	1.456
CIFAR-10 / ConvNet	400k	10^{-3}	10^{-3}	1.028 (65.2)	2	2.201
CIFAR-10 / ConvNet	400k	10^{-3}	10^{-4}	1.044 (65.0)	4	2.330
CIFAR-10 / ConvNet	4M	10^{-4}	10^{-3}	1.025 (69.7)	32	1.081
CIFAR-10 / ConvNet	4M	10^{-4}	10^{-4}	0.831 (72.5)	41	0.905
PTB / char-LSTM	40k	10^{-3}	10^{-4}	2.667 (3.85)	34	2.874
PTB / char-Transformer	40k	10^{-2}	10^{-4}	2.641 (3.81)	48	2.652

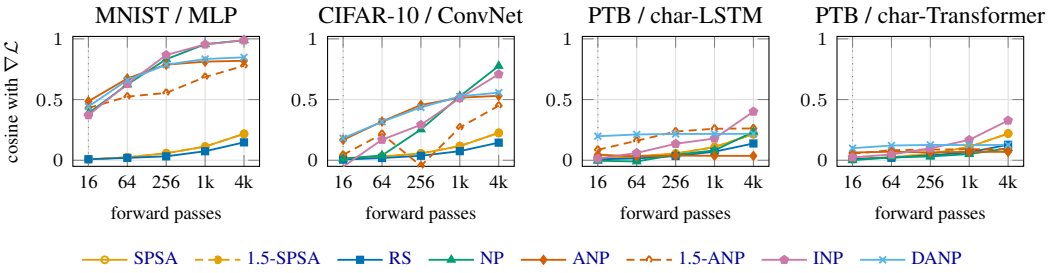


Figure 8: Cosine between each estimator and the true gradient of one batch (whole parameter vector) versus the number of forward passes spent, $\sigma = 10^{-3}$, after a 2000-step backpropagation warm-up. The protocol spends 16–17 passes per step.

that position; the readout), and the position embedding, perturbed once per batch, contributes its row for that position to every sequence and receives the sum over sequences of β_u. The noisy passes run the perturbed recurrent state without the specification's 0.5 damping. $\lambda = 10^{-3}$, $\alpha = 0.5$, $\epsilon_{\text{mach}} = 10^{-12}$.

Language-model specifics. The LSTM is written with a single perturbable layer for the four gate pre-activations so that the gates are the perturbable nodes. The Transformer uses pre-LN blocks with parameter-free layer normalisation, a learned positional embedding over $T = 64$ positions, and a final parameter-free layer normalisation before the readout. The n-gram reference points of Section 5.7 are count models estimated on the training characters and evaluated on the test characters: uniform over $V = 50$, maximum-likelihood unigram and bigram, and a trigram with add-0.1 smoothing.

Forward-forward specifics. One Adam instance updates all parameters; every block's goodness loss is computed on the detached, normalised output of the previous block, so each parameter receives exactly one local gradient per step (the embeddings are trained by the first block's goodness). The label overlay follows Pezeshki [2023]: the first 10 entries of the flattened input are zeroed and the entry of the label is set to the maximum input value (for CIFAR-10, the first 10 pixels of the first row of channel 0); a wrong label is drawn uniformly from the nine other classes; the neutral label sets all 10 entries to 0.1 times the maximum. For PTB, 30 % of the positions of each sequence are replaced by uniformly random characters, resampled every step. Label accumulation sums the goodness over all blocks but the first (Hinton) or over all blocks (logged as a second number). The readout is trained every step by cross-entropy on the normalised, detached top-block activity.

Evaluation and checkpoints. Validation and test metrics are computed on the full splits at initialisation, every minute of training time (every 5 minutes in stage 1) and at the end; the clock is paused

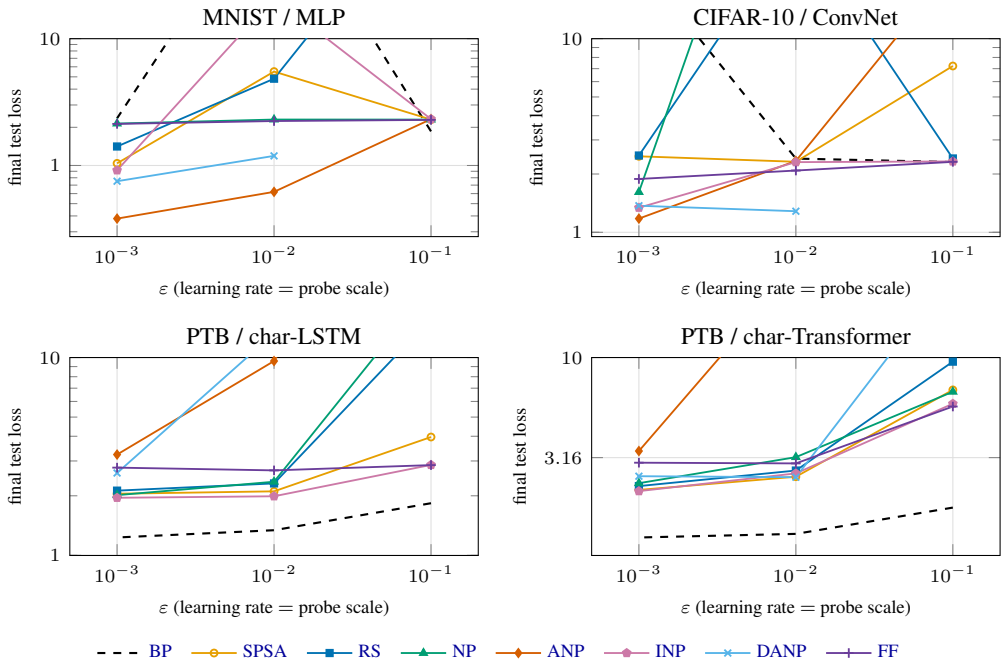


Figure 9: Stage 1: final test loss versus ε (learning rate = probe scale) for the original eight methods; diverged runs are plotted at their last finite evaluation and the y-axis is capped at 10 (values up to 10^6 leave the frame).

during evaluation. Stage 2 saves the parameters at the best validation loss, at 5 minutes and at the end (released with the logs). Runs are not bitwise reproducible from the seed because cuDNN autotuning is on and stopping is time-based.

Hyper-parameters. Table 28 summarises every hyper-parameter; none is tuned per method beyond the ε screening.

Data and licences. MNIST [LeCun et al., 1998], CIFAR-10 [Krizhevsky, 2009] and the character-level Penn Treebank files [Marcus et al., 1993, Mikolov, 2012] are used as distributed by their authors. MNIST is distributed under the Creative Commons Attribution-Share Alike 3.0 licence; CIFAR-10 is released by its authors for research use without a formal licence; the character-level PTB files are the widely redistributed Mikolov preprocessing of the Penn Treebank (LDC99T42) and are used for research only. Our code is released under the MIT licence.

Compute. Stage 1 used about 100 GPU-hours (10.5 hours of wall-clock on the 8-GPU node); stage 2 and the ablations about 500 GPU-hours (24 hours of wall-clock on up to five 8-GPU nodes); the alignment diagnostic takes about an hour on one GPU. Stage 1 was preceded by 32 two-minute smoke runs and four 10-minute backpropagation runs with 30-second evaluations (released with the logs, superseded by Table 5).

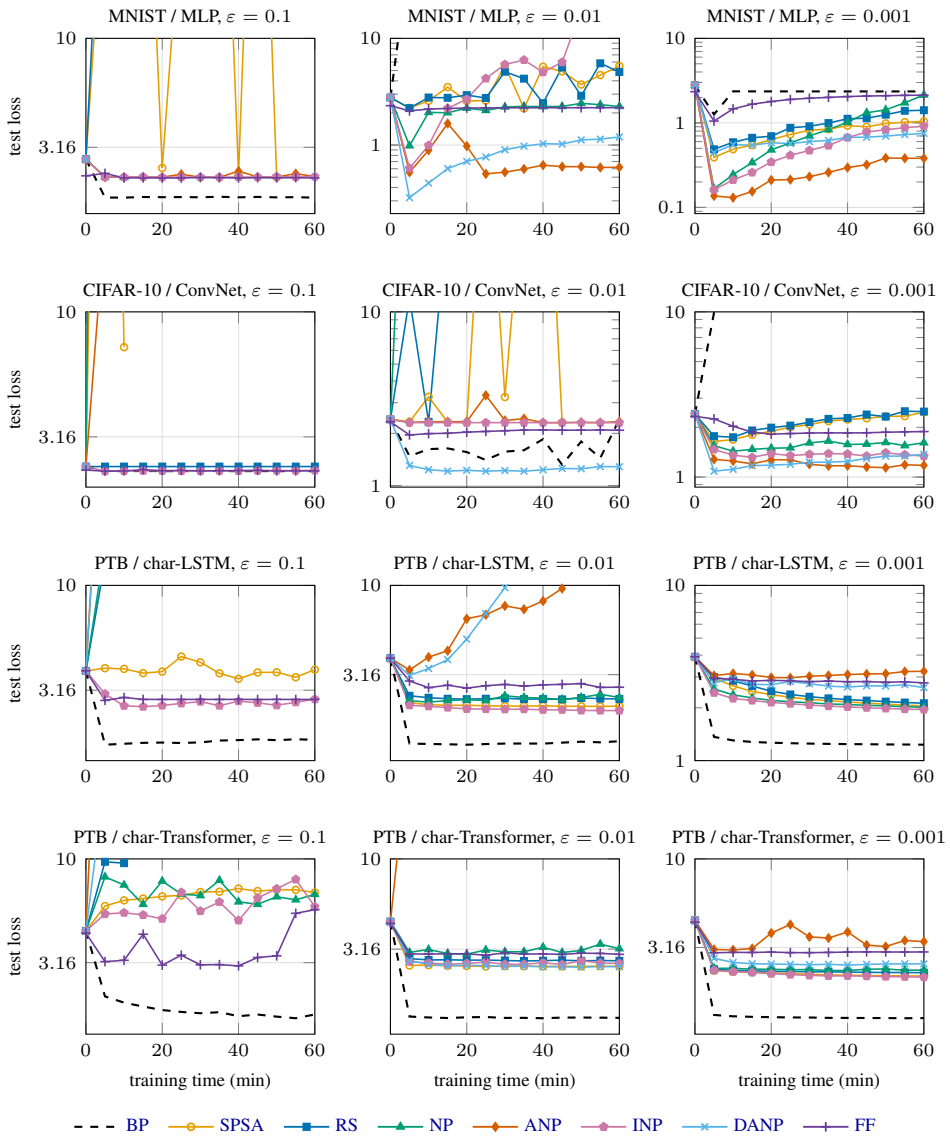


Figure 10: Stage 1: test loss versus training time for every run of the original eight methods. Rows: tasks; columns: $\varepsilon \in \{0.1, 0.01, 0.001\}$.

Table 16: Tied ε screening of every (task, method) pair: the lowest validation loss over the hour at each value (FF on the image tasks: goodness validation accuracy), with the selected value in bold. Values at $\varepsilon \leq 10^{-4}$ come from one-hour runs under the stage-2 settings; the first three columns are stage-1 runs where no stage-2 run exists. Every selected value has a worse neighbour on each side except at a grid edge (10^{-5} for four image-task pairs, 0.1 for FF on MNIST).

Task	Method	$\varepsilon = 0.1$	$\varepsilon = 0.01$	$\varepsilon = 0.001$	$\varepsilon = 0.0001$	$\varepsilon = 10^{-5}$
MNIST / MLP	BP	1.826	2.744	0.314	0.100	**0.094**
	SPSA	2.566	2.087	0.285	**0.132**	0.185
	1.5-SPSA	2.303	2.048	0.287	**0.134**	0.196
	RS	2.744	2.156	0.360	**0.140**	0.206
	NP	2.301	0.752	0.142	0.102	**0.099**
	ANP	2.304	0.446	0.127	0.102	**0.100**
	1.5-ANP	2.304	1.454	0.176	**0.133**	0.140
	INP	2.305	0.396	0.128	**0.098**	0.101
	1.5-INP	–	–	0.123	**0.106**	0.113
	DUST	–	0.231	0.127	**0.109**	0.112
	DANP	2.744	0.242	**0.181**	0.183	–
	FF	**0.952**	0.948	0.918	0.922	–
CIFAR-10 / ConvNet	BP	2.305	1.183	2.395	**0.867**	0.881
	SPSA	2.395	2.303	1.622	**1.268**	1.529
	1.5-SPSA	2.395	2.303	1.586	**1.281**	1.563
	RS	2.395	2.327	1.738	**1.346**	1.591
	NP	2.395	2.395	1.400	**1.059**	1.222
	ANP	2.395	2.303	1.064	**1.022**	1.224
	1.5-ANP	2.395	2.303	1.775	1.538	**1.485**
	INP	2.305	2.302	1.265	**1.052**	1.183
	1.5-INP	–	–	1.434	**1.144**	1.357
	DUST	–	2.302	1.229	**1.122**	1.383
	DANP	2.395	1.129	**1.052**	1.158	–
	FF	0.224	**0.338**	0.334	0.135	–
PTB / char-LSTM	BP	1.765	1.315	**1.268**	1.492	–
	SPSA	3.597	2.117	**2.066**	2.993	–
	1.5-SPSA	3.702	2.102	**2.067**	2.998	–
	RS	3.911	2.314	**2.141**	3.033	–
	NP	3.911	2.239	**2.033**	2.745	–
	ANP	3.911	3.382	**2.970**	2.999	–
	1.5-ANP	2.494	**2.165**	2.287	2.518	–
	INP	2.652	2.007	**1.971**	2.563	–
	1.5-INP	2.124	**1.874**	1.925	2.244	–
	DUST	1.993	**1.747**	1.754	2.074	–
	DANP	3.911	3.151	**2.627**	2.858	–
	FF	2.832	**2.674**	2.789	2.956	–
PTB / char-Transformer	BP	1.731	1.344	**1.299**	1.371	–
	SPSA	4.497	2.554	**2.201**	2.296	–
	1.5-SPSA	4.497	2.561	**2.191**	2.301	–
	RS	4.497	2.728	**2.297**	2.347	–
	NP	4.497	2.957	2.359	**2.353**	2.732
	ANP	4.497	4.497	3.060	**2.721**	2.851
	1.5-ANP	3.330	2.704	2.621	**2.602**	2.701
	INP	4.497	2.613	**2.170**	2.284	–
	1.5-INP	–	2.324	**2.192**	2.241	–
	DUST	2.420	2.313	**1.933**	2.188	–
	DANP	4.497	**2.530**	2.530	2.996	–
	FF	3.046	2.924	2.946	**2.919**	2.946
	IPC	2.281	1.841	**1.829**	1.910	–

Table 17: Stage 1: final validation / test loss of all 96 runs of the original eight methods (seed 0, 5-minute evaluations). Bold: the ε with the lowest *final* validation loss. A dagger marks a diverged run (last finite evaluation shown). Initial validation losses: 2.744 (MNIST), 2.395 (CIFAR-10), 3.911 (LSTM), 4.497 (Transformer).

		validation / test loss		
Task	Optimizer	$\varepsilon = 0.1$	$\varepsilon = 0.01$	$\varepsilon = 0.001$
MNIST / MLP	BP	**1.826 / 1.857**	51.107 / 177.402†	2.239 / 2.355
	SPSA	957.457 / 2.302†	2.193 / 5.502	**0.917 / 1.039**
	RS	52.924 / 170.573†	8.157 / 4.833	**1.238 / 1.412**
	NP	2.301 / 2.301†	2.303 / 2.302	**1.985 / 2.141**
	ANP	2.323 / 2.319	0.519 / 0.619	**0.363 / 0.381**
	INP	2.325 / 2.321	8.917 / 21.010†	**0.935 / 0.916**
	DANP	– / –†	1.033 / 1.190	**0.643 / 0.751**
	FF	2.288 / 2.288	2.243 / 2.241	**2.123 / 2.127**
CIFAR-10 / ConvNet	BP	2.305 / 2.305	**1.357 / 2.402**	23.735 / 24.475†
	SPSA	7.292 / 7.225†	22.978 / 2.313†	**2.300 / 2.468**
	RS	**2.395 / 2.406**	197.361 / 38.724†	2.401 / 2.492
	NP	1.9×10^6 / 1.3×10^6†	314.593 / 326.902†	**1.597 / 1.613**
	ANP	23.474 / 23.143†	2.332 / 2.336	**1.143 / 1.177**
	INP	2.327 / 2.323	2.303 / 2.303	**1.321 / 1.339**
	DANP	– / –†	**1.242 / 1.285**	1.289 / 1.371
	FF	2.311 / 2.311	2.081 / 2.084	**1.880 / 1.886**
PTB / char-LSTM	BP	1.858 / 1.835	1.371 / 1.340	**1.268 / 1.231**
	SPSA	3.991 / 3.965	2.125 / 2.107	**2.066 / 2.048**
	RS	15.802 / 15.689†	2.330 / 2.310	**2.141 / 2.124**
	NP	23.036 / 22.815†	2.380 / 2.356	**2.033 / 2.015**
	ANP	195.280 / 192.326†	9.592 / 9.603†	**3.242 / 3.232**
	INP	2.885 / 2.866	2.007 / 1.989	**1.971 / 1.955**
	DANP	271.941 / 271.567†	13.384 / 13.251†	**2.627 / 2.608**
	FF	2.862 / 2.859	**2.699 / 2.688**	2.789 / 2.776
PTB / char-Transformer	BP	1.802 / 1.780	1.351 / 1.316	**1.300 / 1.262**
	SPSA	6.865 / 6.893	2.557 / 2.539	**2.201 / 2.181**
	RS	9.453 / 9.538†	2.745 / 2.726	**2.297 / 2.279**
	NP	6.617 / 6.746	3.206 / 3.180	**2.374 / 2.357**
	ANP	305.304 / 306.216†	27.783 / 27.601†	**3.433 / 3.412**
	INP	5.921 / 5.880	2.651 / 2.638	**2.170 / 2.151**
	DANP	26.685 / 26.551†	**2.543 / 2.525**	2.564 / 2.554
	FF	5.707 / 5.694	**2.958 / 2.959**	2.976 / 2.983

Table 18: Stage 2: selected ε^*, mean minute of the best validation checkpoint and number of seeds for every (task, method) pair.

	MNIST		CIFAR-10		LSTM		Transformer	
Method	ε^*	min / seeds	ε^*	min / seeds	ε^*	min / seeds	ε^*	min / seeds
BP	1e-05	5 / 3	0.0001	1 / 3	0.001	60 / 3	0.001	57 / 3
SPSA	0.0001	52 / 3	0.0001	60 / 3	0.001	60 / 3	0.001	60 / 3
1.5-SPSA	0.0001	58 / 3	0.0001	59 / 3	0.001	60 / 3	0.001	59 / 3
RS	0.0001	46 / 3	0.0001	59 / 3	0.001	60 / 3	0.001	60 / 3
NP	1e-05	52 / 3	0.0001	59 / 3	0.001	60 / 3	0.0001	60 / 3
ANP	1e-05	56 / 3	0.0001	58 / 3	0.001	40 / 3	0.0001	21 / 3
1.5-ANP	0.0001	29 / 3	1e-05	60 / 3	0.01	10 / 3	0.0001	44 / 3
INP	0.0001	7 / 3	0.0001	56 / 3	0.001	60 / 3	0.001	60 / 3
1.5-INP	0.0001	13 / 3	0.0001	56 / 3	0.01	52 / 3	0.001	59 / 3
DUST	0.0001	15 / 3	0.0001	60 / 3	0.01	49 / 3	0.001	59 / 3
DANP	0.001	1 / 3	0.001	5 / 3	0.001	44 / 3	0.01	40 / 2
FF	0.1	29 / 3	0.01	19 / 3	0.01	18 / 3	0.0001	57 / 2
IPC	–	–	–	–	–	–	0.001	52 / 3

Table 19: The same runs at the end of the hour (secondary, stability metric): test loss and accuracy or bits per character after 60 minutes of training. A dagger marks a pair in which at least one seed was stopped by the divergence rule (its last evaluation is used). BP's MNIST and CIFAR-10 values are its over-confident end-of-hour state (Section 5.2), not a ranking.

	MNIST / MLP		CIFAR-10 / ConvNet		PTB / char-LSTM		PTB / char-Transformer		
Method	loss	acc. (%)	loss	acc. (%)	loss	bits/char	loss	bits/char	score
DUST	0.190±0.010	97.1±0.1	1.119±0.030	60.7±1.2	**1.716±0.010**	**2.48±0.01**	**1.932±0.022**	**2.79±0.03**	4.96
1.5-INP	0.186±0.014	97.2±0.2	1.161±0.025	59.6±0.7	1.878±0.018	2.71±0.03	2.166±0.010	3.13±0.02	5.39
NP	**0.106±0.002**	**96.8±0.1**	**1.035±0.058**	**64.2±2.5**	2.092±0.067	3.02±0.10	2.328±0.011	3.36±0.02	5.56
1.5-SPSA	0.141±0.006	95.9±0.2	1.216±0.047	56.6±1.8	2.054±0.008	2.96±0.01	2.154±0.025	3.11±0.04	5.57
INP	0.317±0.065	96.9±0.2	1.065±0.017	62.9±0.8	2.019±0.056	2.91±0.08	2.176±0.022	3.14±0.03	5.58
SPSA	0.139±0.005	96.0±0.2	1.200±0.050	57.7±1.8	2.141±0.082	3.09±0.12	2.200±0.017	3.17±0.02	5.68
RS	0.161±0.009	95.2±0.2	1.295±0.055	53.5±2.3	2.229±0.092	3.22±0.13	2.276±0.027	3.28±0.04	5.96
1.5-ANP	0.163±0.005	96.2±0.4	1.532±0.041	46.2±1.0	2.368±0.068	3.42±0.10	2.687±0.176	3.88±0.25	6.75
ANP	0.111±0.005	96.6±0.2	1.068±0.012	63.1±0.6	3.116±0.273	4.50±0.39	2.909±0.124	4.20±0.18	7.20
DANP	0.782±0.024	94.2±0.2	1.355±0.026	64.2±0.8	2.674±0.058	3.86±0.08	2.735±0.297	3.95±0.43	7.55
FF	2.298±0.010	57.7±29.9 [10.9]	2.091±0.022	32.1±0.6 [23.0]	2.777±0.084	4.01±0.12	2.943±0.029	4.25±0.04	–
BP	0.819±0.209	96.7±0.2	14.311±2.756	65.6±0.3	1.245±0.012	1.80±0.02	1.276±0.012	1.84±0.02	17.65
IPC	–	–	–	–	–	–	1.820±0.006	2.63±0.01	–

Table 20: Image tasks: test accuracy (%) and test loss at the evaluation with the highest validation *accuracy*, mean over seeds. Best probe method per task in bold.

	MNIST / MLP		CIFAR-10 / ConvNet	
Method	acc. (%)	loss	acc. (%)	loss
BP	97.1±0.2	0.103±0.003	70.0±0.4	0.964±0.053
SPSA	95.8±0.2	0.143±0.009	57.7±1.8	1.203±0.050
1.5-SPSA	95.7±0.2	0.145±0.009	56.4±1.6	1.222±0.044
RS	95.0±0.3	0.164±0.010	53.6±2.1	1.298±0.052
NP	96.8±0.1	0.107±0.003	**65.2±2.7**	1.006±0.065
ANP	96.6±0.2	0.112±0.004	63.4±1.2	1.055±0.029
1.5-ANP	96.0±0.4	0.161±0.007	45.7±0.9	1.540±0.034
INP	97.0±0.1	0.242±0.061	62.9±0.9	1.068±0.030
1.5-INP	**97.2±0.1**	0.142±0.035	60.1±0.4	1.139±0.015
DUST	97.0±0.1	0.124±0.017	60.5±1.4	1.130±0.039
DANP	94.0±0.0	0.490±0.246	63.2±0.8	1.292±0.024
FF	59.7±29.5	2.335±0.058	33.9±0.3	2.007±0.036

Table 21: Throughput: optimizer steps and forward passes per hour (thousands), mean over the seeds of the selected runs, measured with eight runs sharing one host (one BP step is counted as one pass; its backward is not counted).

	MNIST		CIFAR-10		LSTM		Transformer	
Method	steps/h	passes/h	steps/h	passes/h	steps/h	passes/h	steps/h	passes/h
BP	2683.4	2683.4	2023.8	2023.8	91.6	91.6	560.5	560.5
SPSA	1002.7	16043.3	590.4	9446.4	24.6	394.2	165.1	2641.7
1.5-SPSA	850.3	14455.1	511.2	8690.2	33.6	571.8	227.2	3862.4
RS	1075.8	17212.5	609.0	9743.7	24.4	390.6	170.8	2732.5
NP	503.9	8062.5	336.9	5390.0	17.2	274.7	78.3	1252.3
ANP	365.4	5846.5	264.6	4234.1	14.4	229.7	63.5	1016.7
1.5-ANP	502.9	8549.5	255.8	4348.6	23.5	399.0	73.5	1248.8
INP	598.7	9579.0	277.0	4431.8	18.3	292.9	130.2	2083.3
1.5-INP	408.8	6950.0	266.4	4529.6	13.3	226.1	76.0	1291.5
DUST	235.6	3769.2	176.5	2823.7	13.2	210.6	78.1	1248.9
DANP	430.3	6885.3	250.4	4007.0	12.4	198.8	71.7	1146.8
FF	1761.5	4657.4	958.1	2874.4	81.1	162.3	267.2	534.4
IPC	–	–	–	–	–	–	308.1	308.1

Table 22: Steps completed in the hour (thousands) and milliseconds of training time per step, mean over seeds of the selected runs (eight runs sharing one host).

	MNIST		CIFAR-10		LSTM		Transformer	
Method	steps (k)	ms/step	steps (k)	ms/step	steps (k)	ms/step	steps (k)	ms/step
BP	2683(1383)	1.55	2024(761)	2.02	92(49)	46.19	560(399)	8.44
SPSA	1003(312)	3.89	590(173)	6.55	25(8)	156.07	165(47)	22.86
1.5-SPSA	850(264)	4.59	511(149)	7.56	34(1)	107.12	227(2)	15.85
RS	1076(357)	3.68	609(177)	6.34	24(9)	158.92	171(51)	22.19
NP	504(170)	7.63	337(88)	11.28	17(6)	222.80	78(3)	46.05
ANP	365(111)	10.39	265(76)	14.55	14(5)	267.07	64(2)	56.70
1.5-ANP	503(161)	7.81	256(84)	14.98	23(0)	153.44	73(2)	49.04
INP	599(188)	6.54	277(6)	13.00	18(6)	208.25	130(33)	28.71
1.5-INP	409(11)	8.81	266(6)	13.52	13(1)	271.24	76(2)	47.41
DUST	236(10)	15.30	176(3)	20.40	13(0)	273.67	78(0)	46.12
DANP	430(7)	8.37	250(7)	14.38	12(4)	307.48	72(21)	52.55
FF	1762(106)	2.05	958(18)	3.76	81(51)	54.87	267(5)	13.48
IPC	–	–	–	–	–	–	308(5)	11.69

Table 23: Peak GPU memory of the training process (MB): tensors allocated, and reserved by the caching allocator, recorded during the stage-2 runs.

	MNIST		CIFAR-10		LSTM		Transformer	
Method	alloc.	reserved	alloc.	reserved	alloc.	reserved	alloc.	reserved
BP	243	258	1889	1972	127	142	329	436
SPSA	235	258	1889	1972	118	142	321	420
1.5-SPSA	235	258	1889	1972	118	142	321	420
RS	235	258	1889	1972	118	142	321	420
NP	243	258	1889	1972	151	172	369	482
ANP	243	258	1889	1972	151	172	369	490
1.5-ANP	243	258	1889	1972	148	178	369	538
INP	243	258	1889	1972	145	166	359	466
1.5-INP	243	258	1889	1972	150	177	359	510
DUST	243	258	1889	1972	147	204	347	456
DANP	247	278	3625	3894	154	192	373	494
FF	249	278	1937	2006	135	178	353	430
IPC	–	–	–	–	–	–	360	500

Table 24: Whole-model cosine between each estimator and the true gradient at 1024 forward passes (trained state) as a function of the probe scale, fp32 on the GPU. Only 1.5-ANP, whose weights use a second difference, degrades as σ shrinks.

Model	Estimator	$\sigma = 0.1$	$\sigma = 0.01$	$\sigma = 0.001$	$\sigma = 0.0001$	$\sigma = 10^{-5}$
MNIST / MLP	SPSA	0.02	0.11	0.11	0.11	0.11
	RS	0.00	0.07	0.07	0.07	0.07
	NP	0.96	0.96	0.96	0.96	0.96
	ANP	0.81	0.81	0.81	0.81	0.81
	INP	0.95	0.95	0.95	0.95	0.95
	DANP	0.82	0.83	0.83	0.83	0.83
	1.5-SPSA	0.02	0.11	0.11	0.11	0.11
	1.5-ANP	0.78	0.74	0.69	0.54	0.46
CIFAR-10 / ConvNet	SPSA	0.02	0.12	0.12	0.12	0.12
	RS	0.00	0.06	0.08	0.08	0.08
	NP	0.39	0.50	0.53	0.52	0.53
	ANP	0.12	0.22	0.52	0.52	0.52
	INP	0.42	0.54	0.51	0.52	0.51
	DANP	-0.01	0.29	0.53	0.52	0.52
	1.5-SPSA	0.02	0.12	0.12	0.12	0.11
	1.5-ANP	0.49	0.41	0.27	0.25	0.26
PTB / char-LSTM	SPSA	0.02	0.11	0.11	0.11	0.11
	RS	0.00	0.02	0.07	0.07	0.07
	NP	0.07	0.08	0.08	0.08	0.08
	ANP	0.02	0.04	0.04	0.04	0.04
	INP	0.17	0.18	0.18	0.18	0.18
	DANP	0.13	0.22	0.22	0.22	0.22
	1.5-SPSA	0.02	0.11	0.11	0.11	0.10
	1.5-ANP	0.28	0.29	0.26	0.11	0.12
PTB / char-Transformer	SPSA	0.01	0.09	0.11	0.11	0.11
	RS	0.00	0.00	0.07	0.07	0.07
	NP	-0.00	0.03	0.05	0.05	0.05
	ANP	0.01	0.02	0.07	0.07	0.07
	INP	0.02	0.06	0.17	0.18	0.17
	DANP	-0.03	0.08	0.13	0.13	0.13
	1.5-SPSA	0.01	0.09	0.11	0.11	0.11
	1.5-ANP	0.03	0.09	0.09	0.08	0.05

Table 25: Gradient alignment at 40k and 4M parameters on the sequence models (whole-model cosine with the true gradient, $\sigma = 10^{-3}$, trained state).

		16 passes		1024 passes	
Model	Estimator	40k	4M	40k	4M
PTB / char-LSTM	SPSA	0.014	0.002	0.108	0.012
	RS	0.007	0.001	0.071	0.007
	NP	-0.004	0.006	0.081	0.024
	ANP	0.046	0.020	0.037	0.045
	INP	0.014	0.005	0.179	0.049
	DANP	0.198	0.103	0.217	0.123
	1.5-SPSA	0.014	0.002	0.108	0.012
	1.5-ANP	0.086	0.028	0.260	0.132
PTB / char-Transformer	SPSA	0.010	0.001	0.110	0.011
	RS	0.011	0.000	0.066	0.007
	NP	0.000	-0.000	0.051	0.009
	ANP	0.064	0.018	0.070	0.028
	INP	0.024	-0.004	0.168	0.048
	DANP	0.100	0.014	0.125	0.047
	1.5-SPSA	0.010	0.001	0.110	0.011
	1.5-ANP	0.054	0.037	0.091	0.059

Table 26: Per-layer cosine between each estimator and the true gradient at 1024 forward passes, $\sigma = 10^{-3}$, after the 2000-step backpropagation warm-up (layer indices in module order; "all" is the whole parameter vector).

Model	Layer	SPSA	RS	NP	ANP	INP	DANP	1.5-SPSA	1.5-ANP
MNIST / MLP	0	0.08	0.05	0.95	0.94	0.93	0.94	0.08	0.48
	1	0.30	0.23	0.95	0.90	0.96	0.89	0.30	0.78
	2	0.56	0.35	0.98	0.89	0.99	0.93	0.56	0.83
	all	0.11	0.07	0.96	0.81	0.95	0.83	0.11	0.69
CIFAR-10 / ConvNet	0	0.65	0.41	0.56	0.55	0.46	0.44	0.65	-0.10
	1	0.15	0.10	0.43	0.54	0.48	0.54	0.15	0.10
	2	0.10	0.05	0.45	0.62	0.60	0.60	0.10	0.24
	3	0.04	0.04	0.81	0.93	0.93	0.86	0.04	0.87
	4	0.21	0.15	0.94	0.88	0.99	0.85	0.21	0.87
	all	0.12	0.08	0.53	0.52	0.51	0.53	0.12	0.27
PTB / char-LSTM	0	0.12	0.11	0.58	0.58	0.41	0.97	0.12	0.42
	1	0.08	0.05	0.07	0.04	0.16	0.17	0.08	0.21
	2	0.23	0.15	0.13	0.05	0.27	0.49	0.23	0.36
	all	0.11	0.07	0.08	0.04	0.18	0.22	0.11	0.26
PTB / char-Transformer	0	0.21	0.09	0.44	0.45	0.22	0.81	0.21	0.45
	1	0.15	0.10	0.34	0.33	0.12	0.42	0.15	0.20
	2	0.06	-0.02	0.07	0.15	0.12	0.10	0.06	0.18
	3	0.13	0.10	0.04	0.50	0.11	0.35	0.13	0.38
	4	0.11	0.06	0.03	0.20	0.23	0.45	0.11	0.26
	5	0.17	0.12	0.13	0.16	0.18	0.40	0.17	0.15
	6	0.10	0.11	0.00	0.12	0.21	0.39	0.11	0.14
	7	0.10	0.07	-0.01	0.23	0.19	0.54	0.10	0.23
	8	0.03	0.01	0.05	0.20	0.32	0.06	0.03	0.15
	9	0.08	0.02	0.07	0.03	0.16	0.07	0.08	-0.03
	10	0.11	0.08	0.10	0.13	0.15	0.34	0.11	0.09
	11	0.07	0.08	0.07	0.11	0.23	0.46	0.07	0.26
	12	0.04	0.02	-0.02	0.09	0.10	0.27	0.05	0.21
	13	0.07	0.01	0.07	0.20	0.12	0.53	0.08	0.33
	14	0.04	0.01	-0.02	0.16	0.13	0.40	0.04	0.24
	all	0.11	0.07	0.05	0.07	0.17	0.13	0.11	0.09

Table 27: Precision check on the ConvNet: whole-model cosine with the true gradient at 1024 passes in fp32 with TF32 off, fp32 with TF32 on (PyTorch's default for convolutions) and fp64, same probes.

Model	Estimator	σ	fp32	TF32	fp64
CIFAR-10 / ConvNet	NP	0.01	0.442	0.440	0.459
	NP	0.001	0.445	0.420	0.475
	ANP	0.01	0.189	0.189	0.193
	ANP	0.001	0.481	0.479	0.482
	INP	0.01	0.442	0.439	0.477
	INP	0.001	0.425	0.358	0.484
	SPSA	0.01	0.115	0.115	0.108
	SPSA	0.001	0.116	0.116	0.110

Table 28: Hyper-parameters shared by all runs.

Optimizer	Adam, $(\beta_1, \beta_2) = (0.9, 0.99)$, $\epsilon_{\text{Adam}} = 10^{-8}$, no schedule, no weight decay (AdamW 10^{-2} in the BP ablation only)
Learning rate η = probe scale $\sigma = \varepsilon$	$\{0.1, 0.01, 0.001, 0.0001\}$, 10^{-5} on the image tasks (Table 16) (untied grid in Section 5.3; own recipes in Section 5.4)
Probe distribution	$\mathcal{N}(\mathbf{0}, \mathbf{I})$, scaled by σ
Forward passes per step	16 (probe methods), 17 (1.5-SPSA, 1.5-ANP), 3 / 2 (FF), $1 + 1$ backward (BP)
SPSA, 1.5-SPSA pairs / RS candidates	8 / 15
Curvature exponent α / floor λ	1.5-SPSA: 0.1 / 1.0; 1.5-ANP: 0.5 / 10^{-3}
Batch size	128 images; 64 sequences $\times$ 64 characters
Initialisation	Xavier-uniform, zero biases (LSTM forget gate = 1)
FF threshold / PTB corruption	2 (0.25 for the LSTM block) / 30 % of characters
Decorrelation rate η_R (DANP)	10^{-3}, fixed
Training budget	60 min of training time per run, evaluation excluded (30 min in the decoupling ablation)
Evaluation interval	every 1 min (stage 2) / 5 min (stage 1) of training time, and at the end
Divergence rule	non-finite loss, or validation loss $> 10\times$ its step-0 value at two consecutive evaluations (stage 1: $2\times$, plus the training-loss clause)
Precision	fp32, TF32 off (stage 2); PyTorch default, TF32 on for convolutions (stage 1)
Seeds	3 (stage 2 and the replicated untied settings), 1 (stage 1, screenings, own-recipe grids and controls)

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